Paper 2: The Structure of Admissible Physics

Non-Closure, Gauge Origin, Capacity Counting, and the 61-Type Partition

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Reality is the minimum-cost expression of distinction compatible with finite capacity.

Abstract

Paper 1: *The Enforceability of Distinction* derived the quantum skeleton—Hilbert space, the Born rule, gauge symmetry, tensor products, and entropy—from the four structurally necessary components of the *Principle of Least Enforcement Cost* (PLEC): A1 (finite admissibility capacity — upper bound on admissibility), MD (uniform per-test cost floor — positive lower bound), A2 (minimum cost — argmin selection), BW (budget-window richness — non-degeneracy), together with constitutive admissibility semantics (SP, K3, FD1–FD6). PLEC is the canonical variational formulation of the admissibility-of-distinction foundation; it does not replace that foundation. This paper deploys those results.

We begin with the Non-Closure Theorem (L_{nc}): individually admissible admissibility demands, when composed at shared interfaces, generically exceed local budgets. Admissible sets are not closed under composition. This is the engine behind all competition, selection, and structure in the framework. We provide constructive witnesses and quantitative consequences, then show that non-closure *requires* non-abelian gauge structure. Abelian composition is additive ($\Delta = 0$, the (Sep) branch of the Sep/IJC Representation Theorem in Paper 1's Technical Supplement, where joint pairs separate into single-side defenders and the algebra commutes); this is incompatible with non-closure. Non-abelian routing realizes the (IJC) branch of the same theorem, in which the joint defender's codespace rotates into the substrate pool and sustains $\Delta > 0$. The complete exclusion of abelian carriers is proved in Technical Supplement II [17] ($B1'$, §3.3), and Section 3 develops the structural argument establishing that only non-abelian routing can sustain $\Delta > 0$. At any interface whose substrate is richer than its coarse partition lattice, the partition-structure clause of the Representation Theorem (Paper 1 Supplement v8.9 §"Sep/IJC representation theorem") forces the (IJC) branch, so non-abelian gauge structure is not optional but inherited from the substrate-level admissibility geometry.

The cost function is proved unique via Cauchy's functional equation (1821): the only monotone additive cost function with unit normalization is $F(d) = d$. This eliminates a potential free parameter and anchors the Weinberg angle derivation.

The gauge template is then proved unique by classification across the four infinite families and five exceptional compact simple Lie algebras within the declared carrier filters (the bank check tests a 17-representative list; the families are closed by monotonicity): the three carrier requirements derived in Paper 1 [14] (complex ternary, pseudoreal binary, abelian grading)

select $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ as the only viable product structure. Capacity minimality selects $N_c = 3$ within the SM-family template class. The fermion content is derived by complete scan of the 1,680 chiral templates of the declared space under seven filters, with two unconditional representation-exclusion theorems (P1–P2), P3's joint F2+F5 budget exclusion of colorless triplets, and dominance certificates (P4–P5) against the named enlargement directions. The unique minimum among the enumerated candidates is the Standard Model at 45 Weyl fermions. Stated with its premises, this is the central result: given the full input ledger — the template caps and representation tables, the fixed gauge template, the imported generation count (with generation universality of the abelian action: all continuous abelian charges act identically on the three generation copies), the imported one-loop Weyl coefficient, the anomaly and Witten equations, the F3 doublet–singlet content predicate, CPT identification, the F6 non-degeneracy requirement (nonzero quark-doublet hypercharge, $Y_Q \neq 0$ — an independent assumption named at its point of use, in its canonical full-system form), the Weyl-component objective, and the equal-unit channel convention — the Standard-Model template at 45 Weyl fermions is the unique minimum among the enumerated candidates, computed exactly. The generation count and the abelian-factor count enter honestly: $N_{\text{gen}} = 3$ is consumed as an input at its banked ground (the capacity–beta identities), and the abelian sector is closed a-posteriori by the closure theorems given the scanned content and the typed-inventory input (I-typing, whose second clause — the relative-torus matter action — is stated in Supplement I's hypothesis ledger). Technical Supplement I's conditional classification theorem carries the full input ledger (C1–C10); the grounds of the premise list are the object of Technical Supplement II's graded program.

Capacity counting—structural admissibility channels, not field-theoretic degrees of freedom, one equal-weight slot per Weyl component, Higgs component, and gauge generator (the channel-identification convention, Supplement bridge axiom BA5)—yields $C_{\text{total}} = 61$ $(45 + 4 + 12)$. The MECE (mutually exclusive, collectively exhaustive) partition by two mechanism predicates gives:

$$\underbrace{61}_{C_{\text{total}}} = \underbrace{42}_{\text{vacuum}} + \underbrace{3}_{\text{baryonic}} + \underbrace{16}_{\text{dark}}. \tag{1}$$

This partition, proved saturation-independent, identifies the cosmological density fractions $\Omega_\Lambda = 42/61 = 0.6885$ and $\Omega_m = 19/61 = 0.3115$ via horizon equipartition (L_{equip}) — an identification made under the equipartition bridge's explicit assumption set (equal priors at saturation and the horizon identification; stated as bridge axioms BA5–BA7 in Technical Supplement I's bridge-axiom ledger), not as an unconditional consequence of the gauge/matter core.

All results trace to named check functions in the codebase. No continuous free parameters are introduced; the discrete premises of the classification (the scan caps, F6's quark-doublet hypercharge non-degeneracy, generation universality, the admissibility-completeness reading) are named at their points of use. The correlation space now has rooms, and we can count them.

A note on framing. No process of "selection" occurs in this derivation. Under A1, the admissible configuration space is well-defined; under A2, its minimum-cost member is realized. That member is the Standard Model. Alternatives (SU(5), SO(10), E_6) are admissible in the sense that their cost fits below A1's capacity bound; they are not rejected by a filter, they simply have higher cost than SU(3)×SU(2)×U(1), so A2 names SU(3)×SU(2)×U(1) as what is. The verbs "selects," "forces," and "rejects" throughout name logical uniqueness results (arg min over admissible configurations under A2), not physical processes. See the companion note *What Is, Is Minimum-Cost Admissible* for the full descriptive reading.

Contents

1. Introduction: from skeleton to architecture

Paper 1: *The Enforceability of Distinction* derived that finite admissibility capacity (A1), together with the other components of the *Principle of Least Enforcement Cost* (MD, A2, BW) and constitutive admissibility semantics, yields quantum structure—Hilbert space, the Born rule, gauge symmetry, entropy—as mathematical consequences. PLEC is the canonical variational formulation of the admissibility-of-distinction foundation: reality is the minimum-cost expression of distinction compatible with finite capacity. It is expressed through four structurally necessary components—A1 (upper bound on admissibility), MD (positive cost floor), A2 (argmin selection), BW (non-degeneracy)—none derivable from the others, all load-bearing throughout this paper. The Technical Supplement [15] provides the end-to-end proof. That paper built the skeleton: it proved that any universe with finite admissibility resources must have an operator algebra, a state space, a gauge symmetry, and an entropy functional. But a skeleton is not a building. Paper 1 said nothing about which gauge group, which particles, or how many of them. The skeleton is compatible with many possible architectures.

This paper shows that only one architecture fits.

The argument begins with a simple observation about how admissibility works—not in extreme conditions, but always.

Consider a hydrogen atom. Two physical distinctions are being maintained at the same interface. The first is the electron's spin: up or down. Spin is genuinely binary—exactly two states, one admissibility channel, the simplest possible distinction the universe can maintain. The second is the electron's position within the orbital: where in the probability cloud the electron actually is. Position is not binary. It is a continuous observable, enforced through a stack of binary distinctions, each one resolving a finer question about location—is the electron in this half of the orbital? this quarter? this eighth? Each layer in that stack costs capacity to maintain, and the depth of the stack determines how precisely position is resolved. It costs capacity to keep each distinction definite, to prevent it from dissolving into indefinite superposition.

The precise identification of which channels carry spin admissibility and which carry position admissibility depends on spacetime structure that this paper does not yet derive. That accounting is completed in Paper 5: *Spacetime and Gravity*, where the local Lorentz frame channels are named and the hydrogen atom example can be given its full channel-level treatment. What this paper establishes is the mechanism: both tasks draw from the same local admissibility pool, their joint cost exceeds the sum of their individual costs, and the consequence is structural. The starting point is the arena established by Paper 1: the finite orthogonal sector form (T_{sector}, Technical Supplement [15]), which packages the substrate into a finite-dimensional vector space V with block-orthogonal inner product, sector decomposition $V = (\bigoplus_d M_d) \oplus \Pi$, and self-adjoint admissibility projections. Paper 2 builds on this arena.

These two admissibility tasks interfere with each other, and the reason is straightforward in capacity language. Both distinctions draw admissibility from the same pool of channels at the same interface. The interface does not have separate, dedicated infrastructure for spin and separate infrastructure for position—it has a single pool of admissibility channels through which all local distinctions must be routed.

When the interface commits channels to enforcing the spin distinction, it partitions the local admissibility resources into two camps: those consistent with spin-up and those consistent with spin-down. This commitment reshapes the landscape of available channels. The channels that remain for enforcing the position distinction are no longer the same channels that would have been available if spin had not been enforced first. Some channels well-suited for position admissibility have already been committed to the spin partition; the position task must now work with what remains, in a configuration it did not choose.

The result is that the second task is harder than it would have been in isolation—not because capacity has been "used up" in a simple bookkeeping sense, but because the first commitment has rearranged the infrastructure that the second task must operate on. The additional cost of working in this rearranged landscape is the interference surplus, $\Delta > 0$. It is not an overhead fee or a penalty—it is the genuine cost of enforcing a distinction in a context that has already been shaped by another admissibility commitment.

This interference is not a crisis that arises when the budget is tight. It is the normal mode of operation. The surplus Δ is present in a lone hydrogen atom drifting in deep space with vast capacity headroom. It is present in every quantum system, at every scale, in every regime. The Heisenberg uncertainty principle is the most familiar consequence: you cannot simultaneously enforce both spin and position with arbitrary precision because the joint cost always exceeds the sum of the individual costs, regardless of how much capacity is available. The interference is structural—a property of how admissibility works, not of how close the system is to saturation.

What finiteness adds is not the interference itself, but its consequences. In a universe with unlimited capacity, the surplus Δ would be real but harmless—every combination of admissibility tasks would still fit, no matter how much interference they generated. Nothing would be excluded, nothing selected, no structure forced. But capacity is finite (A1). The interference costs accumulate. They eat into a budget that has an upper bound. And this means that not every combination of individually affordable admissibility tasks remains affordable when composed. Two things that each fit within the budget can collectively exceed it—not because something went wrong, but because the interference surplus that was always present now pushes the total past the limit.

This is *non-closure*: the set of individually admissible admissibility configurations is not closed under combination. Putting it plainly: nature can afford A, and nature can afford B, but nature cannot always afford A and B at the same time. Non-closure was proved in Paper 1 [14] (§4.3); this paper explores its consequences.

At the substrate level, the hydrogen atom developed above is a worked instance of the four middle-regime commitments named at *Paper 0 v6.1.0 §sec:substrate_middle_commitments*. The spin and position distinctions are real because the substrate has paid to hold them (commitment C1 — "things are real"); they realign to other admissible alignments at positive cost (commitments C2 + C3 — "motion goes somewhere" and "you have to push hard enough", formalized as Paper 1 supplement v8.40 Definition 3 `def:realignment-cost`); the joint correlation between them carries the substrate's separate charge for holding linked distinctions (commitment C4 — "linking is a different task", the source of the interference surplus $\Delta > 0$). The interference Δ developed below is the C4-commitment fee in concrete physical form; non-closure is what happens when the substrate's middle-regime budget runs out of room for that fee on top of the per-channel costs.

Non-closure has two faces, and the hydrogen atom contains both. The electron shows the mild face: spin and position interfere, the joint cost exceeds the sum ($L_{\text{nc}}^{\text{interf}}$, $\Delta > 0$), and the Heisenberg uncertainty principle is the result. The proton shows the severe face: each quark carries a colour charge, and at hadronic scales the admissibility cost of an isolated coloured state exceeds the available capacity entirely—the configuration is inadmissible. What remains is the colour-singlet proton, the only combination whose cost fits. Confinement is not a force that traps quarks. It is non-closure selecting the only surviving configuration.

Those consequences are far-reaching. Combinations whose total admissibility cost exceeds the available capacity are simply excluded—they overflow the budget and cannot be maintained. But many combinations do fit. Among those satisfying all structural requirements, the combination with the lowest total admissibility cost is the one that persists. This is not a selection principle layered on top of the physics—it *is* the physics: the Principle of Least Enforcement Cost under structural constraints, with costs shaped by A1 and MD, admissibility constraints set by A1, the argmin selection rule supplied by A2, and the non-degeneracy of that argmin guaranteed by BW. The realized configuration is the arg min of total cost on the admissible set. The question becomes: what is cheapest?

To answer that question precisely, we need two concepts that will appear throughout this paper.

The first is **superadditivity**. We have already seen it at work in the hydrogen atom: the joint admissibility cost exceeds the sum of the individual costs by the surplus $\Delta > 0$. The formal statement is $\varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2) + \Delta$, where Δ is the cost of enforcing the second distinction in a channel landscape that has already been rearranged by the first.

This pattern is familiar from everyday experience, even if the word is not. Two roommates sharing a kitchen do not simply sum their individual grocery bills; they must also coordinate schedules, negotiate shelf space, and avoid conflicts over the stove. The coordination overhead is the surplus. Two musicians playing simultaneously do not just add their individual sounds; they must also manage the acoustic interaction between them—the result is not "musician A plus musician B" but something qualitatively different that depends on how their contributions interact. In each case, sharing infrastructure creates overhead that would not exist if the tasks were performed independently.

The second is the distinction between **abelian** and **non-abelian** structure. An abelian system is one where the order of operations does not matter: performing task A then task B gives the same result as B then A. Ordinary addition is abelian: $3 + 5 = 5 + 3$. Depositing money in a bank account is abelian: depositing \$100 then \$50 leaves the same balance as depositing \$50 then \$100. In capacity language, an abelian admissibility commitment does not rearrange the channel landscape—it simply removes capacity from the pool without affecting how the remaining capacity is configured. The second task finds the same infrastructure it would have found anyway, just with less of it. No rearrangement means no surplus: $\Delta = 0$.

A non-abelian system is one where order matters: A-then-B gives a different result from B-then-A. This is not exotic—it is the everyday reality of the physical world. Putting on socks then shoes produces a different result from putting on shoes then socks. Rotating a book 90° around the x-axis then 90° around the z-axis leaves the book in a different orientation than the reverse order—pick up any book and try it. Driving north then turning east puts you in a different place than turning east then driving north. Twisting a Rubik's cube: a clockwise turn of the top face followed by a clockwise turn of the right face scrambles the colors differently than the reverse order, because each twist rearranges the pieces that the next twist acts on.

The pattern is clear: whenever the first operation changes the *context* in which the second operation acts, order matters. In capacity language, this is precisely what we saw in the hydrogen atom: committing admissibility channels to spin rearranges the landscape for position, and vice versa. The rearrangement is asymmetric—committing to spin first and then enforcing position costs a different total than the reverse order—because the two commitments reshape the channel landscape differently. This asymmetry is the operational origin of noncommutativity: $AB \neq BA$. And the physical consequence is the surplus $\Delta > 0$: if performing A rearranges the context for B, then the joint cost of A and B includes the cost of operating in that rearranged context.

This is why the distinction matters for physics. An abelian universe—one where no admissibility commitment rearranges the channel landscape—would have $\Delta = 0$ everywhere. Every combination of distinctions that individually fits would also jointly fit. There would be nothing to select, nothing to exclude, no structure. But the physical world is manifestly non-abelian. The order in which you measure spin and position matters. The order in which you apply color rotations to quarks matters. Non-closure requires $\Delta > 0$; $\Delta > 0$ requires channel rearrangement; channel rearrangement requires non-abelian structure. The gauge group of the universe cannot be abelian because the universe has non-closure, and non-closure demands it.

With these concepts in hand, we derive the unique architecture in seven stations.

Section 2 makes the machinery of non-closure precise—superadditivity, path dependence, and saturation regimes—and shows that these are not three separate phenomena but three faces of a single mechanism. The reader who has followed the hydrogen atom example already understands the physics; Section 2 gives it teeth.

Section 3 is where the framework first makes contact with the Standard Model. The three carrier requirements—ternary, chiral, and abelian grading—emerge not from empirical observation but from the structure of non-closure itself. By the end of Section 3, without ever naming a Lie group, the framework has already determined *what kind of thing* the gauge group must be. The identification of the specific group follows in Section 6, but the constraints that force it are set here.

Section 5 closes the last door through which a free parameter could enter. The gauge derivation requires a cost function that ranks algebras by expense. If that function were a free choice, everything downstream—including the Weinberg angle—would inherit an arbitrary parameter. A 200-year-old theorem of Cauchy proves the cost function is unique: $F(d) = d$, no alternatives, no wiggle room. After Section 5, the framework has zero adjustable parameters.

Section 6 is the main event. The three carrier requirements meet the classification of compact simple Lie algebras—four infinite families and five exceptional cases, tested through a 17-representative list with the families closed in closed form—and every candidate except one is eliminated. $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is not selected from a shortlist; it is the only structure that survives the declared carrier filters. A complete scan of the 1,680 chiral fermion templates of the declared space under seven sequential filters then leaves five feasible classes, of which the Standard Model at 45 Weyl fermions is the unique capacity minimum. A final subsection derives the scalar sector: one complex $\mathrm{SU}(2)$ doublet, four real components, derived from the chiral fermion content and A2 (minimum cost).

Section 7 introduces a counting principle that departs from standard field theory. What A1 constrains is not field-theoretic degrees of freedom—not polarizations, not helicities—but structural admissibility channels: the capacity types that the interface must independently track. The count is $C_{\text{total}} = 61$ — counted in the equal-unit channel convention (one slot per Weyl component, Higgs component, and gauge generator; Supplement BA5) — and it includes a contribution that field theory does not see: the 42 capacity types consumed by the vacuum itself, the admissibility cost of maintaining empty space.

Section 9 partitions those 61 types into three sectors by two binary predicates—gauge addressability and confinement—and proves the partition is exhaustive: no third predicate exists. The result is $42 + 3 + 16 = 61$, with the vacuum sector consuming 69% of the total budget. This partition is proved saturation-independent: the fractions 42/61 and 19/61 are topological invariants, locked by discrete type classifications, not by continuous parameters. A short equipartition argument (L_{equip}) then identifies these capacity fractions with the cosmological density parameters Ω_Λ and Ω_m — under the equipartition bridge's explicit assumption set (Technical Supplement I, bridge axioms BA5–BA7) — matching observation to 0.05%.

Section 11 answers what is perhaps the most famous open question in particle physics: why are there exactly three generations of fermions? The answer is a capacity ladder. Each generation costs admissibility resources, and the cost grows *quadratically* with the number of generations because every new generation must be distinguished from all existing ones. Three generations fit within the available budget. Four do not. The number of generations is not a free parameter—it is the largest integer compatible with the capacity wall.

Section 12 returns to the hydrogen atom with the full machinery in place. The electron's 7 active channels are named and its absent channels identified as structural predictions. The proton's 24 channels are enumerated and confinement is located precisely as non-closure in the SU(3) baryonic sector. The hydrogen atom as a two-body system is then described as the combination of both channel sets. The section closes with an exact statement of what Paper 5 will complete.

Section 14 describes what we now know about the correlation space—the mathematical object whose skeleton was established in Paper 1 [14]. That paper proved the space exists and has the right mathematical type. This paper has filled in its rooms, its walls, and its budget. The correlation space is no longer abstract: it is a 61-dimensional object with $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ symmetry, three generations of fermions, and a vacuum sector that is the most expensive thing in the universe.

One paper, two technical supplements. The proof burden of this paper lives in two companion documents. Technical Supplement I, *The Classification Core* [16], carries the finished classification mathematics on declared inputs: the conditional classification theorem with its full input ledger (C1–C10), the 1,680-template scan and its waterfall, anomaly and hypercharge closure, the one-U(1) a-posteriori closure theorems, the Lie classification within the declared carrier filters, the capacity count, and the bridge-axiom ledger. Technical Supplement II, *The Foundational Gauge Program* [17], carries the chain from the APF axioms toward those inputs, each link at its banked grade with its premises named — the admissibility algebra and gauge identification, the matter category and lift, the EC/CM grounds, the $C(G) = \dim(G)$ cost program — together with a grade ledger and an open-lane register headed by the fixed-point program. This paper is the argument-first guide and carries no proof burden of its own. Claims about the classification cite Supplement I; claims about the foundational chain cite Supplement II.

Descriptive-framing convention (read once, applies throughout)

In the derivation chapters below, the verbs "forces," "selects," "rejects," and "enforces" are used as shorthand for uniqueness and arg min results over the space of A1-admissible configurations. "A1 forces X" means X is the unique admissible value under A1 (i.e., all competitors are excluded from the admissible set). "Capacity minimality selects Y" means Y is the arg min (under A2) of capacity cost over the A1-admissible set. "Z is rejected" means Z is not admissible under the stated filter. No process of admissibility, selection, or rejection is being described. "Enforcement" names the cost structure $\varepsilon(d)$; it does not denote an agent. The shorthand is retained for readability—rewriting sentence by sentence would obscure the proof structure—but the underlying content throughout is a sequence of theorems asserting uniqueness (under A1) or arg min properties (under A2) of the admissible set. See the companion note *What Is, Is Minimum-Cost Admissible* for the full descriptive restatement.

2. Non-closure: the engine of structure

Operational primitives: where each structural term is defined and used

Paper 2 takes Paper 1's quantum skeleton (A1 + PLEC $\to$ finite-dimensional complex Hilbert space) as input and establishes the gauge/matter architecture — the proved unique minimum among the enumerated candidates (Technical Supplement I), resting on the graded foundational program (Technical Supplement II) — with the cosmological fractions as bridge readings (BA5–BA7). Five structural primitives carry the argument. This box is a wayfinder, not new content; every claim points to the section that defines or formalizes the term.

Co-located distinctions. Two distinctions sharing an interface, with disjoint mechanisms but a common substrate. *Defined:* Paper 1 L_Δ §3.1 (imported). *Used throughout:* §2, §3, §6.

Non-closure. The admissibility algebra admits no single representation valid across all regimes: some combinations of individually admissible distinctions are collectively inadmissible. *Formalized:* §2 $L_{\mathrm{nc}}^{\mathrm{budget}}$ (budget overflow) and the finer non-closure of §3 (the algebraic version that drives gauge structure).

Gauge template. The product structure $SU(3) \times SU(2) \times U(1)$ as the unique admissible non-abelian gauge configuration, with $N_c = 3$. *Proved unique:* §6 (Lie classification within the declared carrier filters, family-wise over the declared carriers with a 17-representative tested list), and §6.2 ($N_c = 3$ as the arg min over admissible color-group ranks).

Capacity partition. $C_{\mathrm{total}} = 61 = 42 + 19$, with 42 vacuum-sector channels and 19 matter-sector channels. *Counted:* §7 (per-channel accounting), §9 (MECE vacuum / baryonic / dark split). *Downstream consequences:* $\Omega_\Lambda = 42/61$, $\Omega_m = 19/61$ in §9; $\sin^2\theta_W = 3/13$ in §10; $d_{\mathrm{eff}} = 102$ in §9.4.

Fermion template. The 45-fermion Standard Model content as the unique capacity minimum among the five feasible classes left by the seven filters on the 1,680 gauge-compatible candidates. *Enumerated:* §6 (three-stage argument + complete scan of the declared space). *Hypercharges derived:* §6.

The three structural predictions of §13 ($\theta_{\text{QCD}} = 0$, vacuum stability, proton stability) use these primitives together; no additional input is introduced.

2.1 The theorem and its witnesses

Non-closure was derived in Paper 1 [14] (§4.3) from A1 and L_{loc} (Locality). We restate it here with additional constructive content, beginning with the plain-English statement: things that individually fit within a budget can collectively exceed it.

Lemma $L_{\text{nc}}^{\text{budget}}$ (Budget Non-Closure)

Statement. The set of admissible distinction-sets is not closed under composition. There exist individually admissible admissibility demands E_1, E_2 such that:

$$E_1 \leq C, \qquad E_2 \leq C, \qquad E_1 + E_2 > C. \tag{2}$$

In plain language: nature can afford to enforce E_1 alone, and nature can afford to enforce E_2 alone, but nature cannot afford to enforce both simultaneously at the same interface.

Proof. By L_{loc}, admissibility distributes over interfaces each with finite budget $C(\Gamma)$. By L_{ε^*}, every distinction costs at least $\varepsilon^* > 0$. Any two demands each exceeding $C/2$ overflow when composed. **Witness:** $C = 10$, $E_1 = E_2 = 6$. □

Note. This is a pure budget statement. It holds for additive costs with $\Delta = 0$ and requires no quantum structure. The classical reading is: two items costing \$6 each exceed a \$10 budget when purchased together. [a]

[a] Code reference: `check_L_nc` in `core.py`.

Lemma $L_{\text{nc}}^{\text{interf}}$ (Interference Non-Closure)

Statement. When two distinctions d_1, d_2 are enforced *at the same interface* with a shared channel resource, their joint cost strictly exceeds the sum of their individual costs:

$$\varepsilon(\{d_1, d_2\}) \;=\; \varepsilon(d_1) + \varepsilon(d_2) + \Delta(d_1, d_2), \qquad \Delta > 0. \tag{3}$$

Derivation. Established in Paper 1 via the bridge chain $L_\Delta \to L_{\text{blk}} \to T_{\text{alg}}$ (full proof: Technical Supplement [15], §3–4; Paper 1 Supplement v8.9 §"Sep/IJC representation theorem" fixes the chain at branch (IJC)). Substrate-richness makes branch (IJC) *admissible*, not mandatory: the spectator (Sep) countermodel of the Technical Supplement (the minimal Sep/IJC witness, the $\theta = 0$ case) is a genuine countermodel to any derivation of noncommutativity from A1 alone, so whether an interface presents branch (IJC) records is an empirical occupancy profile read off its correlations, not forced by the axiom; the results here run *given* that branch. The $\Pi \neq \{0\}$ condition is the substrate-side signature of branch (IJC); spectator-substrate pairs (branch (Sep)) yield Π inert, $\Delta = 0$, and a commutative algebra. The pre-algebraic route derives $\Delta > 0$ from K2, K3, T_{sep}, and SP (non-null directions in the pool Π) without invoking Hilbert space or field selection. The key step: coordinating two distinctions at a shared interface requires defending both against pool-supported perturbations—a third admissibility act beyond the two individual ones, whose cost is $\Delta > 0$. The joint defender's codespace W_* is rotated into Π (L_Π), producing the off-diagonal component $F_\Pi \neq 0$ that makes the admissibility algebra noncommutative (T_{alg}). Formally: the automorphism group of the local operator algebra is non-trivial (T3), so the admissibility map is order-sensitive, and the marginal cost $m(d_2 \mid d_1) = \varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) \neq \varepsilon(d_2)$.

Distinction from $L_{\text{nc}}^{\text{budget}}$. Budget non-closure holds even with $\Delta = 0$ (additive costs in a classical world). Interference non-closure is the structural (non-budget) statement: $\Delta > 0$ is a property of the admissibility algebra, not merely a consequence of a tight budget. It is the cost face of the quantum obstruction — necessary for noncommutativity but, on its own, not sufficient for it, since a classically-correlated record can carry $\Delta > 0$ and still admit a global Boolean section. The gauge derivation in Section 3 uses $L_{\text{nc}}^{\text{interf}}$, not $L_{\text{nc}}^{\text{budget}}$. [a]

[a] Code reference: `check_L_nc, check_T1` in `core.py`.

The witness is minimal but the mechanism is universal. For n sectors with demands $E_i > C/n$, the composition exceeds C. This is the structural origin of competition: when finite resources cannot accommodate all admissibility tasks simultaneously, some combinations persist and others are excluded. Among the surviving combinations, the one with the lowest total cost is the architecture that the framework selects.

Non-closure has a reading that ties this paper to the rest of the framework, and it is the reading the title names. To say the admissibility algebra does not close is to say there is no single global description valid across all regimes — no one classical assignment that reproduces every record a set of co-located distinctions presents at once. In the language the quantum sector uses, there is no global Boolean section over the co-available records. That missing section is one obstruction wearing several coats. Read within an interface it is quantum non-commutativity, the object Paper 5 builds Hilbert space and the Born rule from. Read as what a locked colour record costs, it is confinement. Read across separate regions it is the gauge connection the framework declines to instantiate, treated in Paper 12. The superadditive surplus $\Delta > 0$ of $L_{\text{nc}}^{\text{interf}}$ is the *cost* face of this obstruction — necessary for the quantum reading, but not by itself sufficient, since a classically-correlated record can carry $\Delta > 0$ and still admit a global section. The sharp quantum condition is the missing section itself. And whether a given interface realizes it is read off the world, not forced by the axiom: the spectator (Sep) interface is admissible, and A1 draws the line between the two branches without placing any interface on one side.

2.2 Three quantitative consequences of non-closure

Non-closure has three immediate quantitative consequences, each of which plays a structural role in the sections that follow. The first—superadditivity ($L_{\text{nc}}^{\text{interf}}$)—was developed at length in the introduction: joint admissibility at a shared interface costs more than the sum of individual costs by the interference surplus $\Delta > 0$, because the first admissibility commitment rearranges the channel landscape for every subsequent task. We state it formally in the box below and move to the other two.

The second consequence is **path dependence**. Channel rearrangement is not symmetric. When the interface commits to spin first, the channels available for position are rearranged in one way; when it commits to position first, the channels available for spin are rearranged in a different way. The total cost of enforcing spin-then-position is not the same as the total cost of enforcing position-then-spin, because the two orderings produce different rearrangements.

This means there is no "true" cost of enforcing position that exists independently of context. The cost is always relative to the current state of the channel landscape—which commitments have already been made, and how they have reshaped the remaining infrastructure. Each new admissibility task encounters a terrain that has been sculpted by everything that came before it. This is the operational content of noncommutativity: the order of operations matters because each operation reshapes the terrain for whatever comes next.

The third consequence is **saturation**. In the introduction we saw that the interference surplus Δ is always present—it is structural, not crisis-driven. But the *impact* of Δ depends on how much slack remains in the budget.

When demand is low—when only a few distinctions are being enforced and the budget has ample headroom—each new commitment rearranges the channel landscape, but there is enough unused capacity to absorb the interference cost comfortably. This is the quiet regime: Δ exists, but it does not yet force hard choices about what can coexist. Perturbation theory works because the corrections are small relative to the available slack.

As demand increases, the surplus begins to bite. The remaining slack shrinks, and rearrangement costs that were once easily absorbed now compete with new admissibility tasks for the dwindling budget. Distinctions that coexisted peacefully in the quiet regime begin to crowd each other out. This is the competitive regime—the regime of quantum exclusion, decoherence, and the transition from "everything fits" to "something has to give."

Near the capacity limit, the system is maximally constrained. Nearly every channel is committed; any new admissibility task must displace an existing one. The surplus Δ is no longer a correction to the budget—it dominates the budget. This is where confinement lives: the admissibility cost of maintaining a non-singlet colored state exceeds the available slack, and only singlet combinations survive. At the wall itself—full saturation—no slack remains, and the system admits no further distinctions at all.

The admissibility spectrum is not a metaphor. It is a quantitative classification of physical regimes by a single dimensionless ratio: total demand divided by total capacity. The box below formalizes all three consequences.

Consequences of $L_{\mathrm{nc}}^{\mathrm{interf}}$: Superadditivity, Path Dependence, Saturation

Superadditivity. When two distinctions d_1, d_2 share admissibility resources at an interface, their joint admissibility cost exceeds the sum of their individual costs (Eq. 3, $\Delta > 0$). The surplus Δ is what separates quantum admissibility from classical bookkeeping: in a classical budget, costs simply add; in quantum admissibility, coordinating two distinctions at a shared interface requires an additional admissibility act whose cost is $\Delta > 0$.

Path dependence. The marginal cost of enforcing d_2 given that d_1 is already active differs from the cost of enforcing d_2 alone:

$$m(d_2 \mid d_1) \;=\; \varepsilon(\{d_1, d_2\}) - \varepsilon(d_1) \;\neq\; \varepsilon(d_2). \tag{4}$$

Variables:

- $m(d_2 \mid d_1)$ — the marginal cost of enforcing d_2 given that d_1 is already being enforced. Because $\Delta > 0$, this exceeds $\varepsilon(d_2)$.

This is the operational origin of noncommutativity (T1 in Paper 1 [14]): the order in which admissibility commitments are made affects the total cost. Enforcing spin-up-vs-down *first*, then position-here-vs-there, costs a different amount than the reverse order.

Saturation regimes. As the total admissibility demand approaches the capacity, the system enters progressively more constrained regimes. The admissibility spectrum organizes all of physics by the ratio of demand to capacity:

$$\frac{E}{C} \;=\; \frac{\sum_d \varepsilon(d)}{C(\Gamma)} \tag{5}$$

Variables:

- E/C — the saturation ratio. Ranges from 0 (no admissibility demands) to 1 (capacity fully committed).
- *Regimes*: quiet ($E/C \ll 1$, perturbation theory exact), competitive ($E/C \sim 0.3$–0.5, exclusion and decoherence), saturation ($E/C \sim 0.6$–0.9, confinement and phase transitions), wall ($E/C \to 1$, singularities and breakdown).

[a]

[a] Code reference: `check_L_nc, check_L_irr, check_L_irr_uniform` in `core.py`.

3. Why non-closure requires gauge structure

The connection between non-closure and gauge structure is the central structural argument of the framework. The logical chain runs: L_Δ (superadditivity, from K2 + K3 + T_{sep} + SP) derives $\Delta > 0$ at any interface with a non-trivial pool—branch (IJC) of the Sep/IJC Representation Theorem (Paper 1 Supplement v8.9 §"Sep/IJC representation theorem"; spectator-substrate pairs land in branch (Sep) and admit no $\Delta > 0$). L_{blk} shows that direct-sum encoding of jointly enforced distinctions is impossible when $\Delta > 0$. The joint defender's codespace rotates into the pool (L_Π), producing off-diagonal generators $F_\Pi \neq 0$ that make the admissibility algebra noncommutative (T_{alg}: $[E_{d_1}, F_\Pi] \neq 0$). The Hilbert space is then constructed from the algebra via GNS (T_{GNS}), not assumed beforehand. Non-closure is not a consequence of non-commutativity; non-commutativity is a consequence of non-closure. At the substrate level (*Paper 0 v6.1.0 §sec:chained_boats* + Paper 5 supplement v6.7 Proposition `measurement-as-class-transition`), the (IJC) branch is the substrate-state at which the four middle-regime commitments fire non-trivially — the interface-cost commitment C4 ("linking is a different task") produces the surplus $\Delta > 0$ that drives the rest of the chain. The (Sep) branch is the substrate-state at which $\kappa_{\Gamma,\mathrm{int}}$ collapses and the algebra closes commutatively. The (Sep)/(IJC) dichotomy is therefore a refinement of the four middle-regime commitments at the per-pair level, not an independent piece of structure; the gauge derivation that follows runs through the (IJC) branch because that is where the substrate's middle-regime commitment to interface cost has structural consequences. The Technical Supplement [15] provides the complete derivation; the introduction sketched the intuition; we now trace the consequences for gauge structure. Substrate-richness makes branch (IJC) *admissible*, not mandatory — the spectator (Sep) interface is a genuine countermodel, so whether a generic interface presents (IJC) records is an empirical occupancy profile, not forced by the axiom (Technical Supplement, the minimal Sep/IJC witness). The gauge derivation does not need it forced everywhere: it runs *given* the coherent (IJC) regime — the same premise under which Paper 5 derives the quantum skeleton this paper takes as input — so it adds no separate IJC postulate but inherits one, and the group is structural given that regime.

3.1 Abelian composition cannot produce non-closure

If admissibility costs were purely additive—if $\varepsilon(\{d_1, d_2\}) = \varepsilon(d_1) + \varepsilon(d_2)$ with $\Delta = 0$ everywhere—then any pair of individually admissible distinctions fitting within half the budget would compose admissibly. There would be no superadditivity, no path dependence, no noncommutativity. The admissibility algebra would be abelian (all operations commute), the dynamics classical, and the structure trivial.

An abelian gauge group—a product of U(1) factors, where U(1) is the group of phase rotations—routes admissibility additively. Each U(1) channel contributes independently; costs simply add. This is the additive countermodel of Paper 1 [14] (§4.3): $\Delta = 0$ implies full reversibility and no non-closure.

3.2 Non-abelian routing is derived

$L_{\mathrm{nc}}^{\mathrm{interf}}$ requires $\Delta > 0$: admissibility costs at shared interfaces are superadditive (Eq. 3). The key step is that $\Delta > 0$ directly implies non-commutativity of the admissibility algebra — not merely "channel rearrangement" in some vague sense, but a provable operator identity. The argument runs through two named lemmas: L_{irred} establishes that the admissibility algebra is irreducible, and $L_{\Delta \to \mathrm{nc}}$ then uses Schur's lemma to derive non-commutativity.

Lemma L_{irred} (Irreducibility of the Derived Algebra)

Statement. The local admissibility algebra acts irreducibly on the GNS Hilbert space $\mathcal{H}_\omega$ (Theorem T_{GNS}, Technical Supplement [15]). That is, $\mathcal{H}_\omega$ has no proper closed subspace that is invariant under every admissibility operator.

Proof. Suppose for contradiction that a proper closed invariant subspace $\mathcal{V} \subsetneq \mathcal{H}_\omega$ exists. Every state in $\mathcal{V}$ and every state in $\mathcal{H}_\omega \setminus \mathcal{V}$ are mapped, under every admissibility operator, into their respective subspaces. This means the admissibility algebra cannot produce any operator whose action distinguishes a state in $\mathcal{V}$ from a state outside it. Equivalently: two states, one from $\mathcal{V}$ and one from $\mathcal{H}_\omega \setminus \mathcal{V}$ that agree on all operators within $\mathcal{V}$, are assigned identical admissibility records. But A1 requires that all physically distinct admissible states receive distinguishable admissibility records. These two states are distinct (they lie in different subspaces) yet receive the same record: a direct contradiction. Therefore no proper invariant subspace exists; the algebra acts irreducibly. □

Logical status. Irreducibility is a *consequence* of the derived algebraic structure, not a premise in the derivation of noncommutativity. The primary noncommutativity proof (T_{alg}, Technical Supplement) is pre-Hilbert: it derives $[E_{d_1}, F_\Pi] \neq 0$ from L_Δ, L_{blk}, and L_Π without invoking Hilbert space or Schur's lemma. L_{irred} then confirms that the algebra, once represented on $\mathcal{H}_\omega$, is irreducible—a consistency check, not a load-bearing step. [a]

[a]Code reference: `check_L_irr` in `core.py`.

Lemma $L_{\Delta\to\text{nc}}$ (Interference Non-Closure Implies Non-Commutativity)

Statement. The admissibility algebra of the admissibility framework is non-commutative: there exist admissibility operations A, B such that $AB \neq BA$.

Proof (primary route: pre-Hilbert, from Technical Supplement [15]). Three steps.

Step 1 (L_Δ: superadditivity). For co-located distinctions d_1, d_2 at an interface Γ with non-trivial pool $\Pi \neq \{0\}$ (the (IJC) branch of the Sep/IJC Representation Theorem; the alternative (Sep) branch has Π inert and yields $\Delta = 0$ classical structure), pool-supported perturbations threaten both distinctions simultaneously. K2 + K3 + SP give $\Delta(d_1, d_2) > 0$: the joint admissibility cost strictly exceeds the sum.

Step 2 (L_{blk} + L_Π: codespace rotation). Direct-sum encoding ($E_{d_1} + E_{d_2}$) achieves cost $\varepsilon(d_1) + \varepsilon(d_2)$ but fails to defend against pool perturbations. The minimum-cost joint defender π_{W_*} has a codespace W_* rotated into Π by angle $\theta \neq 0$. The off-diagonal component $F_\Pi := \pi_{W_*} - E_{d_1} - E_{d_2} \neq 0$.

Step 3 (T_{alg}: commutator witness). For $v \in M_{d_1}$: $[E_{d_1}, F_\Pi]\, v = -\pi_v \neq 0$, where $\pi_v \in \Pi$ is the pool component of $\pi_{W_*} v$. The full algebra $\mathcal{A} = \text{Alg}_{\mathbb{R}}(\{E_d\} \cup \{E_{\{d_1,d_2\}}\})$ admits no faithful commutative $*$-representation. □

Corollary (Schur route). Once the algebra is represented on $\mathcal{H}_\omega$ (via T_{GNS}), L_{irred} confirms irreducibility, and Schur's lemma gives the familiar characterization: any operator commuting with the entire irreducible algebra must be a scalar. Since admissibility operators are non-trivial, non-commutativity follows independently.

Connection to $\Delta_{12} \neq \Delta_{21}$. The non-commutativity $AB \neq BA$ implies that the composition cost is order-sensitive. Define Δ_{12} as the surplus when d_1 is committed before d_2, and Δ_{21} as the surplus in the reverse order. Since $AB \neq BA$, the states $AB|\psi\rangle$ and $BA|\psi\rangle$ are distinct, encoding different admissibility records. The surplus terms Δ_{12} and Δ_{21} measure the additional cost encoded in these distinct records; since the records differ, $\Delta_{12} \neq \Delta_{21}$ for some pair (d_1, d_2).

Distinction from classical contention. A classical resource-contention system can produce $\varepsilon(\{d_1, d_2\}) > \varepsilon(d_1) + \varepsilon(d_2)$, but with a symmetric surplus $\Delta_{12} = \Delta_{21}$. That matches $L_{\text{nc}}^{\text{budget}}$

(classical pigeonhole). The pre-Hilbert route derives asymmetry ($\Delta_{12} \neq \Delta_{21}$) directly from codespace rotation (L_Π): the rotation angle θ generically differs depending on admissibility order. Witness: $\sigma_x\sigma_z = -\sigma_z\sigma_x$ (`check_T0`, core.py). [a]

[a]Code reference: `check_T0, check_L_nc` in `core.py`.

Formally, T3 (Paper 1) then establishes that the automorphism group of the non-commutative local admissibility algebra is the gauge group. If the algebra were abelian, every automorphism would be trivial (phase rotations only), and $\Delta = 0$ identically. Non-closure requires a non-trivial gauge group with non-commuting generators.

Gauge identification (detailed in Technical Supplement II [17], §2). Technical Supplement II proves the gauge identification theorem: every continuous physical gauge redundancy is inner — $\mathrm{Gau}_0 \subseteq \mathrm{Inn}(\mathcal{A}) = \prod_k \mathrm{PU}(n_k)$ unconditionally, with equality under a named full-invariance hypothesis — sitting in the exact sequence $1 \to Z(\mathcal{A}) \to \mathcal{U}(\mathcal{A}) \to \mathrm{Inn}(\mathcal{A}) \to 1$. Abelian factors arise exclusively from the center $Z(\mathcal{A})$ and carry no self-interaction ($\Delta = 0$ within the abelian sector). For the Standard Model template, the lift from $\mathrm{PU}(n)$ to $\mathrm{SU}(n)$ and the global structure $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y]/\mathbb{Z}_6$ are established as derived consequences.

The precise non-abelian criterion. Interference non-closure ($L_{\mathrm{nc}}^{\mathrm{interf}}$) is not merely superadditivity ($\Delta > 0$) but *order-asymmetric* superadditivity ($\Delta_{12} \neq \Delta_{21}$). A classical congestion system can have $\Delta > 0$ with $\Delta_{12} = \Delta_{21}$ (the contention cost is symmetric in the order of operations). Only when $\Delta_{12} \neq \Delta_{21}$—which $L_{\Delta\to\mathrm{nc}}$ derives from codespace rotation (L_Π) in the admissibility algebra—does the algebra become genuinely non-commutative. It is therefore order-asymmetric superadditivity, not mere superadditivity, that requires non-abelian gauge structure. [1]

Proof boundary. The argument above establishes that the gauge group must be non-trivial and non-commutative. A complete proof that *no* abelian routing can sustain $\Delta > 0$ requires Lemma $B1'$ (Technical Supplement II [17], §3.3), which shows that any complex carrier admitting only bilinear invariants produces composites satisfying $B = \bar{B}$—the composite is self-conjugate and provides no independent admissibility channel under T_M. The ternary invariant and complex carrier required by R1 are the *minimal* structure that survives this constraint. Paper 2 establishes the structural necessity of non-abelian routing via the oriented-composite argument; $B1'$ closes the proof that bilinear (abelian-type) carriers are generically excluded.

3.3 What non-abelian structure creates

Non-abelian routing solves the non-closure problem—it produces the interference surplus $\Delta > 0$ that the framework requires. But it creates new problems that the framework must also solve.

In an abelian gauge group like U(1), the admissibility channels are neutral—they direct traffic but do not participate in it. A photon routes electromagnetic admissibility between charged particles, but the photon itself carries no electric charge. In a non-abelian gauge group, this is no longer true. The admissibility channels themselves carry charge. They interact with each other. The traffic cops are themselves subject to traffic law.

[1]Code reference: `check_T3` in `core.py`.

This self-interaction is not optional—it is the mathematical signature of non-commutativity. If the generators of the group do not commute, their commutator produces new generators, and those generators act on each other. The admissibility infrastructure is entangled with itself. This self-entanglement has consequences that cascade through the framework, ultimately determining the specific gauge group and matter content of the universe.

To follow the cascade, we need to understand four things: why the gauge group must be a *Lie group*, why the strength of gauge interactions changes with energy scale, why it changes in the specific direction it does, and why that direction leads to confinement.

3.4 Why the gauge group is a Lie group

The admissibility algebra operates on a continuous state space—this was derived in Paper 1 [14] (T2: Hilbert space via GNS, Technical Supplement [15] T_{GNS}). The symmetries of a continuous space form a continuous group. A continuous group with a smooth structure is a *Lie group*, named after the mathematician Sophus Lie. Discrete symmetry groups—like the group of permutations, where you can swap objects but only in whole jumps—cannot capture the full symmetry of a continuous admissibility structure. They would leave gaps: directions in the state space that no symmetry operation reaches. The gauge group must be a Lie group because the state space it acts on is continuous.

Among Lie groups, the relevant ones are *compact*—meaning, roughly, that the group has finite volume. This is required by A1: an infinite-volume gauge group would require infinite capacity to distinguish its elements, since admissibility must resolve *which* group element is active at each interface, and distinguishing points in a space of infinite volume requires infinitely many independent measurements. Compact Lie groups are the finite-capacity-compatible symmetries.

The classification of compact simple Lie algebras is a finished problem in mathematics: four infinite families (A_n, B_n, C_n, D_n) and five exceptional cases (G_2, F_4, E_6, E_7, E_8) [11]. The elimination below works family-wise — closed-form monotonicity handles each infinite family, and the bank check tests a 17-representative list spanning the low-rank members and all five exceptionals. This finite list is what makes the complete elimination over the declared carriers in Section 6 possible: the framework does not search an infinite landscape, but checks every entry on a short, complete list against the declared carrier filters.

3.5 Why coupling constants change with energy

That coupling constants "run" with energy—that the effective strength of an interaction depends on the scale at which it is measured—is one of the most precisely tested facts in quantum field theory. The discovery that non-abelian couplings grow at low energy (asymptotic freedom) was made independently by Gross and Wilczek [2] and by Politzer [3], and has been confirmed experimentally across many orders of magnitude in energy [4]. The admissibility framework does not discover this phenomenon; it *re-derives* it from A1 without importing any field-theoretic machinery, and the capacity-language explanation makes the mechanism transparent.

The strength of a gauge interaction—how strongly the admissibility channels influence the distinctions they route—is not a fixed number. It depends on the resolution at which you probe the system.

The reason is that admissibility channels are not bare wires. They are embedded in a medium of virtual fluctuations—other admissibility commitments that exist at every scale, briefly borrowing and returning capacity. When you probe the system at high energy (fine resolution, short distances), you see the channels before the surrounding fluctuations have had time to accumulate. The effective coupling is relatively clean. When you probe at low energy (coarse resolution, long distances), you see the channels dressed by all the fluctuations between your probe scale and the channel itself. The accumulated fluctuations modify the effective coupling strength.

This is not a quantum mystery—it is a consequence of the same channel-rearrangement logic we saw in the introduction. Each virtual fluctuation is an admissibility commitment that reshapes the channel landscape. The coupling you measure at any given scale reflects the cumulative rearrangement produced by all fluctuations between that scale and the bare channel.

Which direction the modification goes—whether the effective coupling grows or shrinks at low energy—depends on what the fluctuations are doing. And this is where the distinction between abelian and non-abelian becomes consequential once more.

3.6 Asymptotic freedom: why coupling grows at low energy

In an abelian gauge theory (like electromagnetism), the virtual fluctuations are matter-field pairs—electron-positron pairs that briefly pop into existence and screen the central charge, the way a dielectric medium screens an embedded charge. The screening makes the effective coupling *weaker* at long distances. This is why the electromagnetic coupling constant is small at everyday energies: you are seeing a screened version of a larger bare coupling.

In a non-abelian gauge theory, something additional happens. The admissibility channels carry charge—they interact with each other. Their self-interaction produces an *anti-screening* effect: instead of diluting the coupling, the gauge-channel fluctuations amplify it. For non-abelian groups with sufficiently few matter fields, the anti-screening from gauge self-interaction overwhelms the screening from matter fluctuations. The net effect is that the coupling grows at low energy and shrinks at high energy.

This is *asymptotic freedom*: the interaction becomes weaker at short distances (high energy) and stronger at long distances (low energy). In the admissibility framework, it is derived rather than imported: L_{AF} follows from the fact that non-abelian admissibility channels carry charge (a consequence of non-commutativity) and that the resulting self-interaction anti-screens (a consequence of the sign structure of the gauge algebra). The derivation requires no input beyond the non-abelian structure that non-closure already derived.

3.7 Confinement: finite capacity meets growing coupling

Now the cascade is complete. At low energy, the non-abelian coupling grows. As it grows, the cost of maintaining a state that carries non-abelian charge increases—because the strongly-coupled admissibility channels demand more and more capacity to keep the charged state distinguished from the vacuum.

In a universe with unlimited capacity, this would be no problem. The coupling could grow without bound, the cost of charged states could grow without bound, and they would all remain admissible. But capacity is finite (A1). There is a scale—the IR saturation scale—where the admissibility cost of maintaining a non-singlet state exceeds the available capacity at the interface. Beyond that scale, non-singlet states are simply inadmissible. They are not "confined by a force" in the Newtonian sense—they are excluded by the budget. Only gauge-singlet combinations, whose net non-abelian charge is zero, can be maintained within the finite capacity.

This is $T_{\text{confinement}}$: confinement is derived from A1 + asymptotic freedom + finite capacity. The β-function sign is verified with the imported one-loop coefficient in Technical Supplement I [16], §2, with the capacity-side sign heuristic in Technical Supplement II [17], §4; the quantitative one-loop coefficient is derived in Paper 4A. The mechanism is not imported from lattice QCD or any other external theory. It follows directly from the cascade:

non-closure → non-abelian structure → self-interacting channels → asymptotic freedom → IR coupling growth → capacity overflow → only singlets survive.

[2]

3.8 Three independent admissibility problems

With confinement established, three independent admissibility problems become visible—each arising from a different aspect of what non-closure does, each requiring its own carrier. A *carrier* is the mathematical object that tracks how admissibility obligations transform under the gauge group; it is the representation in which matter fields live.

The confinement problem. Confinement produces stable singlet composites—objects whose total non-abelian charge is zero. But for those composites to provide independent admissibility channels (as required by T_M, admissibility independence), they must be *oriented*: a composite must be robustly distinguishable from its anti-composite. Unoriented composites—those where a composite and its conjugate can be identified by an internal relabeling—provide no independent admissibility channel, because the "forward" and "backward" versions collapse into the same physical state.

What does orientation require of the carrier? Each constraint eliminates alternatives:

- *Complex type* ($V \ncong V^*$). If the carrier were real or pseudoreal ($V \cong V^*$), there would exist an equivariant antilinear map $J^{\otimes n}$ identifying every composite with its conjugate: $B \leftrightarrow \bar{B}$. The oriented distinction would collapse. Complex type blocks this identification—conjugation is non-trivial, and composites are robustly distinguishable from anti-composites.
- *Three-dimensional.* Orientation requires a genuinely trilinear invariant $\epsilon_{ijk}v^i w^j x^k$—one that cannot be decomposed into pairwise contractions. Two-dimensional carriers admit only bilinear invariants ($v^i w_i$), which are effectively abelian: they contract pairs, not triples, and provide no oriented composites. Four-dimensional and higher carriers are non-minimal by Schur–Weyl duality: they contain the trilinear structure but add unnecessary capacity cost. Three is the minimum dimension that supports irreducible trilinear contraction.
- *Faithful.* Every non-identity group element must act non-trivially on the carrier. If some group elements acted trivially, the carrier would fail to distinguish them, and admissibility could not resolve the full gauge structure. No group information can be lost.

Nothing else satisfies all three constraints simultaneously. The confinement problem requires a carrier that is complex, three-dimensional, and faithful—**R1**.

[2] Code reference: `check_T_confinement, check_T_particle` in `gauge.py`.

The irreversibility problem. Non-closure requires irreversibility (L_{irr}, derived in Paper 1, §4.5): some admissibility commitments cannot be undone. But admissibility independence (T_M) requires that this irreversibility be *intrinsic* to the gauge sector—each gauge factor must provide its own irreversible channels, not merely inherit irreversibility from gravity or from the environment. This is a stronger requirement than simply producing entropy: a vector-like gauge theory coupled to an environment can produce entropy through decoherence without the gauge sector itself containing any intrinsically irreversible process. The requirement is that the gauge structure, in isolation, breaks time-reversal symmetry.

The question is: what makes a gauge theory intrinsically irreversible? To answer this, consider what makes a gauge theory *reversible*. A theory that treats left-handed and right-handed fields identically is called "vector-like." Such a theory has four properties that, taken together, guarantee zero intrinsic irreversibility:

- Every interaction vertex has a CPT-conjugate vertex that exactly reverses it.
- Gauge-invariant bare masses exist without spontaneous symmetry breaking—particles can be massive without any mechanism to "break" the symmetry.
- No sphalerons arise—there are no topologically irreversible processes.
- All CP-violating phases can be rotated away by field redefinitions.

A vector-like theory is, at the gauge level, a perfectly reversible machine. Every process has an exact inverse. This violates T_M: the gauge sector provides zero irreversible channels of its own.

The only escape is *chirality*: the gauge theory must treat left-handed and right-handed fields *differently*. Once left and right are treated asymmetrically, the four reversibility conditions above all fail: CPT conjugation no longer maps every vertex to its reverse, bare masses are forbidden (requiring a Higgs mechanism), sphalerons become possible, and irremovable CP phases appear. The gauge sector acquires intrinsic irreversibility.

What carrier provides chirality at minimum cost?

- *Pseudoreal type.* The carrier must be orientation-asymmetric—it must distinguish left from right. Real carriers ($J^2 = +1$) are orientation-blind. Complex carriers work but are non-minimal for this purpose (they provide more structure than chirality alone requires). Pseudoreal carriers ($J^2 = -1$) are the minimal orientation-asymmetric option: they distinguish handedness without the full machinery of complex type.
- *Two-dimensional.* Dimension 2 is the minimum faithful dimension for a pseudoreal carrier. A one-dimensional pseudoreal carrier does not exist ($J^2 = -1$ requires at least a two-dimensional space).
- *Faithful.* As with R1: every group element must act non-trivially.

Chirality at minimum cost requires a carrier that is pseudoreal, two-dimensional, and faithful—**R2**.

The distinguishability problem. The two non-abelian carriers above are powerful but incomplete. They cannot, by themselves, distinguish all physically different states.

To see why, consider what the non-abelian factors actually track. The R1 carrier (complex, 3-dimensional) assigns a "color" label to each state—it tracks which of three color charges the state carries. The R2 carrier (pseudoreal, 2-dimensional) assigns a "weak" label—it tracks whether the state is in a weak doublet or singlet. But some physically distinct states carry the *same* color and weak labels. For example, up-type and down-type antiquarks both transform as $(\bar{3}, 1)$ under color and weak isospin—the non-abelian factors see them identically. Yet they are physically different: they have different masses, different electric charges, and different roles in nuclear physics.

The framework requires that all physically distinct states be distinguishable by the gauge structure — admissibility completeness, a completeness reading of the gauge labels that is motivated by A1 and named as a premise (the type-inventory reading; Technical Supplement II [17]). This requires a third carrier that "breaks the tie":

- *Abelian* (U(1)). The bookkeeping carrier must be abelian—its only job is to assign a distinct numerical label (hypercharge) to each multiplet. A non-abelian bookkeeper would introduce unnecessary self-interaction and capacity cost.
- *Single factor*. One U(1) factor suffices to resolve all degeneracies in the non-abelian spectrum. A second U(1) would be non-minimal under A2 (minimum cost): it would add capacity cost without resolving any additional degeneracy.
- *Distinct charge assignments*. Every physical multiplet must receive a different hypercharge, so that no two representations are conflated by the full gauge structure.

Distinguishability at minimum cost requires a single U(1) factor with distinct charge assignments—**R3**.

These three problems are independent: confinement addresses the stability and orientation of composites, chirality addresses the arrow of time within the gauge sector, and grading addresses the completeness of the labeling system. Each demands its own carrier. Theorem R, derived in Paper 1 [14] (Appendix A), consolidates these three requirements into a single formal statement.

Theorem R: Representation Requirements from Admissibility

Statement. Any admissible interaction theory satisfying A1 must support three independent carriers:

R1 L_{ternary} (Ternary Carrier): A faithful complex 3-dimensional carrier with irreducible trilinear invariant.

Variables and source:

- *Faithful* — every non-identity group element acts non-trivially on the carrier. No group information is lost.
- *Complex* — the carrier V is not isomorphic to its conjugate V^*. This prevents the identification of composites with their anticomposites ($B \neq \bar{B}$).
- *Trilinear invariant* — an irreducible contraction $\epsilon_{ijk} v^i w^j x^k$ that cannot be decomposed into pairwise contractions. This is the mathematical signature of three-body composites (baryons).
- *Source*: L_{nc} (non-closure requires stable composites) + $T_{\text{confinement}}$ (IR saturation makes non-singlets inadmissible; imported-coefficient β-sign check: Technical Supplement I [16], §2) + T_M (admissibility independence requires oriented composites) + B_1' (complex type blocks $B \leftrightarrow \bar{B}$ identification; Technical Supplement II [17], §3.3).

R2 L_{chiral} (Chiral Carrier): A faithful pseudoreal 2-dimensional carrier.

Variables and source:

- *Pseudoreal* — the carrier admits an anti-unitary map $J : V \to V$ with $J^2 = -1$. This is orientation-asymmetric: it distinguishes left-handed from right-handed without being fully complex.
- *Chiral* — the carrier treats left-handed and right-handed fields differently. A non-chiral ("vector-like") theory would be CPT-symmetric at the gauge level, with zero intrinsic irreversibility.
- *Source*: L_{irr} (irreversibility) + T_M (each gauge factor must provide its own irreversible channels).

R3 L_{grading} (Abelian Grading): A single U(1) factor with distinct charge assignments for all physical multiplets.

Variables and source:

- *Abelian grading* — a U(1) "bookkeeper" that assigns a numerical label (hypercharge) to each matter field. Without it, the non-abelian factors cannot distinguish physically distinct states: u^c and d^c both carry the same color and weak charges, differing only in hypercharge.
- *Source*: admissibility completeness under the type-inventory reading (the gauge structure distinguishes all antecedently distinct matter types — A1-motivated, not A1-derived; the reading and its entry-typed inventory are named as premises in Technical Supplement II [17]) + A2 minimality (one U(1) resolves all degeneracies; more than one is non-minimal).

[a]

[a] Code reference: `check_Theorem_R` in `gauge.py`.

These three requirements are independent and jointly exhaustive: R1 addresses non-closure (the confining sector), R2 addresses irreversibility (the chiral sector), and R3 addresses distinguishability (the bookkeeping sector). No reference to any specific Lie group has been made—the requirements are stated entirely in terms of carrier properties.

The next two sections show that these requirements, combined with Lie group classification and capacity optimization, select a unique gauge template and matter content.

4. Confinement and irreversibility are one lock

The confinement problem and the irreversibility problem were just treated as two requirements driving two carriers. They are, at bottom, one mechanism, and seeing why settles a question the lattice treatment of Yang–Mills leaves open: whether a matter-free non-abelian theory confines, with a mass gap, or instead flows to a scale-free, gapless phase.

Start from the surplus. A non-abelian shared-interface commitment is superadditive, $\Delta > 0$ (Eq. 3); the abelian case is additive, $\Delta = 0$. Now add irreversibility. A superadditive commitment is committed at both interfaces at once, and freeing it would take a coordinated act at both — which no single local observer can perform. So it cannot be undone locally. It is permanent: a locked record (L_{irr}). The additive case commits nothing it cannot recover and locks nothing. This is the same split, read twice. $\Delta = 0$ is reversible, abelian, Coulomb, massless; $\Delta > 0$ is irreversible, non-abelian, confining, gapped. Electromagnetism sits on the first line for a reason.

A locked record costs energy to carry, because realignment cost is energy (Paper 3). A permanent commitment of size Δ is a permanent energy of at least Δ, and — once the coloured states have been excluded and only singlets remain — it shows up in the singlet spectrum as a gap. The flux tube that binds a quark is that record; the mass gap is what it costs to hold.

This is what closes the gapless alternative. One might worry that a non-abelian theory could still be scale-free and gapless — and some are: $\mathcal{N} = 4$ super-Yang–Mills, or gauge theory with enough flavours, flows to a conformal fixed point. The reason is matter. Matter supplies additive, reversible, screening channels — the abelian kind — that can net out the gauge surplus and leave the theory reversible. A matter-free theory has none. Every channel is gluonic, every gluonic commitment is $\Delta > 0$, and positive commitments add rather than cancel. With nothing to net them against, the net surplus is positive, the record locks, and the gapless phase — which would need reversible channels to reach — is unreachable. [3]

What this settles and what it leaves open should be said plainly. It settles the confining-versus-conformal question for matter-free non-abelian gauge: the conformal phase is inadmissible, the theory confines, the singlet spectrum is gapped. It does not construct the continuum theory. The infinite-volume and continuum limits, and the reconstruction of a Wightman or Osterwalder–Schrader theory on $\mathbb{R}^4$, are a separate matter, carried as far as present technique allows by the lattice trilogy (Papers 29–31). The grade is accordingly structural: the closure rests on identifying the gapless phase with reversibility, and the matter-free sector with the absence of reversible channels, neither of which reduces to non-closure alone. The lattice papers prove the gap on the lattice; this section is why it is there.

5. The uniqueness of the cost function

Before deriving the gauge template, we close a potential gap: the cost function $F(d)$ that maps the number of admissibility channels d to the per-channel cost. If F were a free function, downstream results (including the Weinberg angle) would depend on an arbitrary choice. We show it is unique.

Lemma L_{Cauchy} (Cost Function Uniqueness)
Statement. The per-channel admissibility cost function $F : \mathbb{N} \to \mathbb{R}^+$ is uniquely determined to be $F(n) = n$ by three conditions:

$$\text{C1 (Additivity):} \quad F(a+b) = F(a) + F(b) \quad \text{for all } a, b \in \mathbb{N} \tag{6}$$

$$\text{C2 (Monotonicity):} \quad a > b \;\Rightarrow\; F(a) > F(b) \tag{7}$$

$$\text{C3 (Unit):} \quad F(1) = 1 \tag{8}$$

Derivation of conditions from A1:

- C1 $\leftarrow L_{\text{loc}} + L_{\text{cost}}$: independently admissible channels at separate interfaces add costs (locality requires additive decomposition over interfaces).

[3] Code reference: `check_T_ym_conformal_phase_excluded_by_record_locking` in `apf/yang_mills_md_bridge.py`.

- C2 $\leftarrow$ A1: more admissibility channels require more capacity (monotonicity of cost in the number of channels).
- C3 $\leftarrow$ Convention: unit choice, analogous to $c = 1$ in relativity.

Proof. Enforcement channels are discrete countable units (A1: capacity is finite and positive, L_{ε^*}: each distinction costs exactly one unit at saturation). The domain of F is therefore $\mathbb{N}$.

Step 1 (Natural numbers). By C1, $F(n) = F(1 + 1 + \cdots + 1) = n \cdot F(1)$ for all $n \in \mathbb{N}$. Applying C3: $F(n) = n$.

Step 2 (Rational extension). Downstream results (the Weinberg angle derivation in Paper 4A) require evaluating F at rational arguments p/q. By C1 applied to q copies of $F(p/q)$: $q \cdot F(p/q) = F(q \cdot (p/q)) = F(p) = p$, giving $F(p/q) = p/q$ for all positive rationals p/q. This is not an additional assumption—it is C1 evaluated on rational multiples. □

Consequence. No downstream quantity in Paper 2 requires the extended domain. The unique result $F(n) = n$ on $\mathbb{N}$ establishes that admissibility cost is linear in channel count, with no free parameter. The full derivation of the trace ratio $\gamma_2/\gamma_1 = d + 1/d = 17/4$ and hence $\sin^2\theta_W = 3/13$ uses the rational extension and is carried out in Paper 4A (T27d), which defines γ_2/γ_1 as the ratio of two sector-level cost functionals and derives both the two-term structure and the addition rule from L_{irr} and R3 (refinement covariance). [a]

[a] Code reference: `check_L_Cauchy_uniqueness` in `L_Cauchy_uniqueness.py`.

The significance of this result is methodological: the uniqueness of $F(n) = n$ eliminates all free parameters from the cost functional. The derivation rests only on three conditions—additivity, monotonicity, and a unit choice—all derived from A1 and L_{loc}. No representation-theory input is required at this stage.

Note on the Weinberg angle preview. Section 10 quotes $\sin^2\theta_W = 3/13 = 0.2308$ and gives a partial derivation. The full logical chain requires the trace ratio $\gamma_2/\gamma_1 = 17/4$, which depends on how the cost functional is applied across two sectors simultaneously. The complete derivation is in Paper 4A; Section 10 states the result with a forward reference so the reader can see where the framework is heading.

6. The gauge template and matter content

6.1 Template uniqueness: Lie classification within the declared carrier filters

Theorem R provides three abstract carrier requirements (R1–R3). We now show that these requirements, applied to the classification of compact simple Lie algebras, select a unique gauge template.

Lemma $L_{\text{gauge,unique}}$ (Gauge Template Uniqueness)

Statement. Given R1–R3 from Theorem R, the gauge group must factor as:

$$G \;=\; \mathrm{SU}(N_c)\times\mathrm{SU}(2)\times\mathrm{U}(1),\qquad N_c\geq 3\ \text{(odd)}. \tag{9}$$

This template is unique. No alternative Lie group structure satisfies all three requirements.

Proof. Six steps.

Step 1 (Factorization). R1 (confining carrier) and R2 (chiral carrier) serve independent admissibility roles: confinement redistributes capacity among incompatible channels (L_{nc}), while chirality distinguishes forward/backward transitions (L_{irr}). By T_M (Monogamy) and L_{loc} (Locality), independent admissibility channels cannot share gauge resources. The gauge group factors as $G_{\mathrm{conf}}\times G_{\mathrm{chir}}\times G_{\mathrm{abel}}$.

Step 2 (Confining factor = $\mathrm{SU}(N_c)$, $N_c\geq 3$). R1 requires a compact simple Lie group whose fundamental representation is faithful, complex, and admits a totally antisymmetric invariant of some rank $k\geq 2$. The minimum rank for oriented composites (Section 3, T_M) is $k=3$: a bilinear invariant ($k=2$, common to real and pseudoreal representations such as $\mathrm{SO}(N)$ and $\mathrm{Sp}(N)$) pairs two states but produces self-conjugate composites ($B=\bar{B}$) that cannot serve as independent admissibility channels. A rank-k antisymmetric invariant on a complex N_c-dimensional carrier has $k=N_c$ (the invariant $\epsilon_{i_1\cdots i_{N_c}}$ of $\mathrm{SU}(N_c)$). Therefore $k\geq 3$ requires $N_c\geq 3$. The family selection is based on which Lie algebra families admit a complex fundamental with any totally antisymmetric invariant; minimality ($k=N_c=3$, i.e., $A_2=\mathrm{SU}(3)$) is applied in the next subsection. Classification over the compact simple Lie algebras (family-wise; the 17-representative tested list), within the declared carrier filters:

Family	**Fund. type**	**R1 verdict**	**Outcome**
$A_n=\mathrm{SU}(n{+}1)$, $n\geq 2$	Complex	Yes (rank $=n+1\geq 3$)	**Pass**
$B_n=\mathrm{SO}(2n{+}1)$	Real	No (real; no complex carrier)	Excluded
$C_n=\mathrm{Sp}(2n)$	Pseudoreal	No (pseudoreal; $B=\bar{B}$)	Excluded
$D_n=\mathrm{SO}(2n)$	Real	No (real; no complex carrier)	Excluded
G_2,F_4,E_8	Real	No	Excluded
E_7	Pseudoreal	No	Excluded
E_6	Complex	Min. faithful dim. = 27 violates A1	Excluded

Only the A_n family ($\mathrm{SU}(N_c)$ with $N_c\geq 3$) passes all R1 requirements. Minimality selects $N_c=3$ in the next subsection.

Step 3 (Chiral factor = $\mathrm{SU}(2)$). R2 requires a faithful pseudoreal 2-dimensional representation. At the Lie algebra level, $\mathfrak{su}(2)$ is the unique compact simple Lie algebra admitting a 2-dimensional irreducible representation. At the group level, the gauge factor must be the simply connected cover $\mathrm{SU}(2)$ rather than its quotient $\mathrm{SO}(3)$: the 2-dimensional (doublet) representation of $\mathfrak{su}(2)$ does not descend to $\mathrm{SO}(3)$, so faithfulness requires $\mathrm{SU}(2)$. All other compact simple Lie algebras have minimum faithful dimension ≥ 3.

Step 4 (Abelian factor = $\mathrm{U}(1)$). R3 requires at least one abelian grading, and U(1) is the unique connected compact 1-dimensional abelian Lie group. That exactly one such factor closes the template is the a-posteriori closure proved in Technical Supplement I [16] for the scanned matter content, under the typed-inventory input (I-typing, including the relative-torus matter action stated in its hypothesis ledger).

Step 5 (Even N_c excluded). Even values of N_c produce zero chiral fermion templates satisfying all admissibility constraints. This is established during the fermion scan (T_{field}, Section 6, filter 5): for even N_c, no template survives the Witten anomaly filter [6], which requires the total number of SU(2) doublets to be even. The result is recorded here: only odd $N_c \geq 3$ remain as candidates.

Step 6 (No simple-group alternative). Any simple group G_{simple} containing $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ has $\dim(G_{\text{simple}}) \geq 24 > 12 = \dim(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))$. The product structure is always cheaper. Independence of admissibility roles (Step 1) also requires factorization. □ [a]

[a]Code reference: `check_L_gauge_template_uniqueness` in `gauge.py`.

6.2 $N_c = 3$ is the unique minimum-cost admissible color group

Step 2 of $L_{\text{gauge,unique}}$ left $N_c \geq 3$ as candidates. Minimality now selects $N_c = 3$. The antisymmetric invariant of $\mathrm{SU}(N_c)$ has rank $k = N_c$; the minimum rank compatible with oriented composites is $k = 3$ (bilinear $k = 2$ is excluded, higher ranks are non-minimal by A2). Therefore $k = N_c = 3$ and the confining factor is SU(3). The table below confirms this via the independent capacity-cost route:

Within the $\mathrm{SU}(N_c)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ template, the anomaly cancellation equation for each N_c is:

$$z^2 - 2z - (N_c^2 - 1) \;=\; 0, \qquad z = Y_u/Y_Q. \tag{10}$$

This is solved analytically for every candidate: $z = 1 \pm N_c$. All odd $N_c \geq 3$ have viable solutions. The capacity cost of each is $\dim(G) = N_c^2 - 1 + 3 + 1 = N_c^2 + 3$. Capacity minimality (A2) selects the cheapest:

N_c	Witten	Anomaly	$\dim(G)$	Cost	Verdict
2	Fail	—	—	—	Excluded (Witten)
3	Pass	$z \in \{4, -2\}$	12	**12**	**Winner**
4	Fail	—	—	—	Excluded (Witten)
5	Pass	$z \in \{6, -4\}$	28	28	Non-minimal
6	Fail	—	—	—	Excluded (Witten)
7	Pass	$z \in \{8, -6\}$	52	52	Non-minimal

$\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ is the unique capacity-optimal gauge group satisfying all admissibility requirements. [4]

6.3 The three-stage argument

The gauge group derivation rests on three logically independent stages, each of which can be attacked separately:

[4]Code reference: `check_T_gauge, check_T4` in `gauge.py`.

Stage 1: Physical necessity (A1 $\to$ carrier requirements R1–R3, Theorem R in Section 3). Finite admissibility determines which information carriers must exist. This stage is pure constraint logic: it uses A1, the structural lemmas, and no Lie theory.

Stage 2: Mathematical classification (R1–R3 $\to$ $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, T_{gauge} in Section 6). The representation theory of compact Lie groups determines that the unique minimal algebra hosting all three carriers is $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)$. This stage is standard mathematics: classification over the compact simple Lie algebras (families closed by monotonicity; a 17-representative tested list), within the declared carrier filters.

Stage 3: Cost closure (A1 + MD $\to$ L_{cost} unique cost functional $\dim(G) \cdot \varepsilon$; A2 $\arg\min \to N_c = 3$ selected; L_{Cauchy} in Section 5). The cost functional measuring "complexity" of a gauge algebra is uniquely determined by L_{cost} (from A1 + MD): $C(G) = \dim(G) \cdot \varepsilon$. A2 then names the $\arg\min$ over viable algebras as the realized configuration. No alternative cost functional compatible with A1 + MD exists. This closes the selection to a theorem rather than a modeling choice.

The independence of the three stages is methodologically significant. A critic can challenge any stage without affecting the others: Stage 1 by exhibiting an A1-compatible theory without R1–R3; Stage 2 by finding a smaller algebra hosting the same carriers; Stage 3 by constructing an alternative A1-compatible cost functional that ranks a different algebra as minimal.

6.4 Fermion content: complete scan of the declared space

The gauge group is fixed. The number of generations is fixed (Section 11). One question remains: which fermions?

The answer is not selected—it is eliminated. The gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ determines the menu of available representations: the mathematical objects that describe how fermions can transform under the gauge symmetry. The question is which combination of representations is consistent with all the constraints that A1 imposes—not one constraint, but seven simultaneously.

The scan is a complete enumeration of the declared space. A *chiral fermion template* is a finite multiset $\mathcal{T} = \{(r_i, n_i)\}$ where each r_i is an irreducible representation of $\mathrm{SU}(3) \times \mathrm{SU}(2)$ and $n_i \geq 1$ is its multiplicity, subject to three constraints: (i) each r_i is a left-handed Weyl field (the template convention; filter 3 below is a doublet–singlet content cut, and the winner's chirality is verified separately in the release's audit layer, not by any filter), (ii) each SU(3) factor lies in $\{3, \bar{3}, 6, \bar{6}, 8\}$ (the AF bound P1 excludes larger representations), and (iii) each SU(2) factor is $\{1, 2\}$ (P2 excludes higher colored representations unconditionally; colorless triplets are excluded a priori by P3's joint F2+F5 budget argument). The template size is bounded by the declared enumeration caps (one colored doublet, at most three colored singlets, one lepton doublet, at most two lepton singlets), so $|\mathcal{T}| \leq 7$; P5 certifies that extra types cannot lower the minimum (dominance against the exhibited 45-DOF survivor — the enumeration's own survivor list contains 6- and 7-entry templates at 48–72 DOF). Under these constraints the search space is finite: 1,680 distinct candidate templates.[5] Five

[5] Colored singlets are unordered within a template (the multiset $\{u^c, d^c\}$ is one template, not two), so the correct enumeration uses combinations with replacement. The canonical standalone verification script reproduces the full scan with exact rational arithmetic, six built-in red-team challenges (RT1–RT6), and both hypercharge solutions; its v2 predecessor is archived at `https://doi.org/10.5281/zenodo.19154197` with an interactive Colab walkthrough at `https://colab.research.google.com/drive/1otO1Qwp_Lr3OsK6ykkeMRATanecPkKg6`. Source code: `https://github.com/Ethan-Brooke/APF-Paper-2-The-Structure-of-Admissible-Physics`.

closed-form proofs (P1–P5 in the box below) bound the space: P1–P2 exclude the enumerated representation enlargements unconditionally, P3 excludes colorless triplets a priori (F2+F5 jointly), and P4–P5 certify dominance against the named extension directions. The caps themselves are declared scan inputs, and their completeness is an open proof obligation of the reproducibility contract (Technical Supplement I [16]); the licensed claim is uniqueness of the minimum among the enumerated candidates.

Seven filters are then applied sequentially. Each filter is an exact declared predicate, typed in Technical Supplement I's input ledger (C1–C10): asymptotic freedom for both gauge factors (the coupling must weaken at short distances; the one-loop coefficient is imported), doublet–singlet content (colored doublets and singlets co-present — a content cut; the winner's chirality is verified separately in the release's audit layer, not by any filter), the anomaly cancellation conditions (the quantum theory must be mathematically self-consistent; imported equations, solved exactly, with the $Y_Q \neq 0$ non-degeneracy an independent assumption in its canonical full-system form), and capacity minimality (among the survivors, the cheapest wins). The filters are not redundant—each eliminates candidates that passed all previous filters.

Of the 1,680 templates, 1,679 fail. One survives.

Theorem T_{field} (SM Fermion Content)

Statement. The unique minimal chiral fermion content consistent with $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, three generations, the F6 non-degeneracy requirement ($Y_Q \neq 0$ — an independent assumption, named at its point of use in the scan), and all remaining admissibility constraints is the Standard Model:

$$\{Q(3,2),\ L(1,2),\ u^c(\bar{3},1),\ d^c(\bar{3},1),\ e^c(1,1)\} \times 3 \ \text{ generations} \tag{11}$$

with $15 \times 3 = 45$ Weyl degrees of freedom.

Proof. Two phases.

Phase 1 (Scan). 1,680 distinct chiral templates constructed from SU(3) representations $\{3, \bar{3}, 6, \bar{6}, 8\}$ crossed with SU(2) representations $\{1, 2\}$, up to 7 field types. Seven sequential filters applied: (1) SU(3) asymptotic freedom, (2) SU(2) asymptotic freedom, (3) doublet–singlet content, (4) $[\mathrm{SU}(3)]^3$ anomaly cancellation, (5) Witten anomaly freedom [6], (6) full anomaly system with non-degenerate rational hypercharge solutions (F6 — an independent assumption requiring nonzero quark-doublet hypercharge, in its canonical full-system form; spectator reduction and uniform dimension are the robustness variants), (7) CPT quotient. Waterfall: 1,680 $\to$ 348 $\to$ 324 $\to$ 24 $\to$ 12 $\to$ 10 $\to$ 5 (post-F6 and post-CPT counts under the canonical full-system predicate, charged-octet solutions included; robustness: the spectator-reduction predicate gives 8 $\to$ 4 and the uniform-dimension code path 4 $\to$ 2 — the minimality winner is identical under all three; Technical Supplement I [16]). Minimality selects the unique survivor at 45 Weyl DOF.

Phase 2 (Exclusion proofs). Five closed-form proofs exclude all categories outside the scan. The asymptotic freedom filters (P1, P2) use the one-loop gauge beta-function coefficients for SU(3) and SU(2); the standard formula is $b = (11/3)C_A - (4/3)T_R N_f$ for N_f Dirac flavors, where C_A is the adjoint Casimir and T_R is the index of the matter representation [1];

the scan's templates are left-handed Weyl fields, for which the matter coefficient is 2/3 per Weyl entry (the convention the verification script has always used). These coefficients are structural results derived in Paper 4A via $L_{\text{beta,capacity}}$ and used here as inputs. No perturbative QFT is imported.

- P1: SU(3) reps ≥ 10 are AF-excluded (single field exceeds budget: $15 > 11$).
- P2: Colored SU(2) reps ≥ 3 are AF-excluded ($12 > 7.3$).
- P3: Colorless SU(2) triplets are excluded a priori by F2+F5 jointly ($3 + 4 + 1 = 8 > 22/3$).
- P4: Two-colored-doublet templates have DOF $\geq 54 > 45$ (class dominance; the minimum is the conjugate-pair template, leptons not required).
- P5: extra field types beyond the anomaly-closed core cannot lower the minimum (dominance; the survivor list's 6- and 7-entry templates sit at 48–72 DOF).

Hypercharges are uniquely determined by anomaly cancellation ($L_{\text{hypercharge}}$, next subsection). □ [a]; standalone verification: `fermion_scan_standalone.py` [18]

[a] Code reference: `check_T_field, check_T5` in `gauge.py`.

That survivor is the Standard Model: three generations of quarks and leptons in precisely the representations and with precisely the hypercharges that experiments observe. The derivation starts from the Principle of Least Enforcement Cost—expressed through its four components (A1 finite capacity, MD positive floor, A2 argmin, BW non-degeneracy)—and arrives here by elimination, through a chain of derivations with no continuous free parameters and with its discrete premises — the scan caps, F6's quark-doublet hypercharge non-degeneracy, generation universality, the admissibility-completeness reading — named where they are used.

6.5 Hypercharges: the numbers textbooks insert by hand

The surviving template has five multiplet types: $Q(3,2)$, $L(1,2)$, $u^c(\bar{3},1)$, $d^c(\bar{3},1)$, $e^c(1,1)$. Each must be assigned a hypercharge Y — the U(1) quantum number from R3 — and the assignments must satisfy the anomaly cancellation conditions that ensure the quantum theory is mathematically self-consistent. There are five unknowns (Y_Q, Y_L, Y_u, Y_d, Y_e) and four independent anomaly conditions. One overall normalization is a convention (the "size" of the U(1) charge unit). So the question is: after fixing the convention, are the hypercharges determined?

Three of the four anomaly conditions are linear and can be solved immediately:

- $[\mathrm{SU}(2)]^2[\mathrm{U}(1)] = 0$: the sum of hypercharges weighted by weak isospin must vanish. This gives $Y_L = -N_c Y_Q = -3Y_Q$.
- $[\mathrm{SU}(3)]^2[\mathrm{U}(1)] = 0$: the sum of hypercharges weighted by color charge must vanish. This gives $Y_d = 2Y_Q - Y_u$.
- $[\mathrm{grav}]^2[\mathrm{U}(1)] = 0$: the sum of all hypercharges (gravitational anomaly) must vanish. After substituting the first two results, this gives $Y_e = -6Y_Q$.

Three unknowns eliminated. Two remain: Y_Q itself (which sets the normalization) and the ratio $z \equiv Y_u/Y_Q$ (which carries the physical content).

The fourth condition is the cubic anomaly $[\mathrm{U}(1)]^3 = 0$: the sum of cubed hypercharges must vanish.

This looks formidable — it involves fifth powers and cross-terms. But after substituting the three linear solutions, the entire expression simplifies. Define $z = Y_u/Y_Q$ and substitute $Y_L = -3Y_Q$, $Y_d = (2-z)Y_Q$, $Y_e = -6Y_Q$ into $[\mathrm{U}(1)]^3 = 0$:

$$2N_cY_Q^3 + 2Y_L^3 - N_cY_u^3 - N_cY_d^3 - Y_e^3 = 0. \tag{12}$$

Every term is proportional to Y_Q^3. Dividing through and collecting terms:

$$z^2 - 2z - 8 = 0. \tag{13}$$

The cubic anomaly condition has reduced to a quadratic. The discriminant is $\Delta = 4 + 32 = 36$, giving two rational roots: $z = 4$ and $z = -2$. These are related by the $u \leftrightarrow d$ exchange ($z \to 2 - z$): they describe the same physics with the up-type and down-type labels swapped. There is one physical solution.

$L_{\text{hypercharge}}$ **(Unique Hypercharge Assignment)**

Statement. Given the surviving template $\{Q(3,2),\ L(1,2),\ u^c(\bar{3},1),\ d^c(\bar{3},1),\ e^c(1,1)\}$ with $N_c = 3$, anomaly cancellation uniquely determines the hypercharge ratios. Normalizing $Y_Q = 1/6$ (conventional):

$$Y_Q = \tfrac{1}{6}, \quad Y_u = \tfrac{2}{3}, \quad Y_d = -\tfrac{1}{3}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = -1. \tag{14}$$

Derived consequences:

- *Electric charges* ($Q = T_3 + Y$): $Q(u) = 2/3$, $Q(d) = -1/3$, $Q(e) = -1$, $Q(\nu) = 0$.
- *Charge quantization*: the anomaly structure fixes all hypercharge *ratios* (a unique rational ray, up to $u \leftrightarrow d$); with the compact-U(1) unit convention $Y_Q = 1/6$, all electric charges land on integer multiples of 1/3. The ratios are derived; the unit is the convention.
- *Quark-lepton relation*: $Y_L = -N_c Y_Q$ ties the lepton hypercharge to the number of colors. The connection between quarks and leptons is not a feature of grand unification imposed from above — it is derived from anomaly cancellation within the derived gauge group.

Proof. Four anomaly conditions on five unknowns. Three linear conditions fix Y_L, Y_d, Y_e in terms of Y_Q and $z = Y_u/Y_Q$. The cubic $[\mathrm{U}(1)]^3 = 0$ reduces to $z^2 - 2z - 8 = 0$ (Eq. 13), with roots $z = 4$ and $z = -2$ related by $u \leftrightarrow d$. One normalization fixes $Y_Q = 1/6$. □ [a]

[a] Code reference: `check_L_anomaly_free` in `gauge.py`.

Every number in the Standard Model's hypercharge column — the numbers that have been measured in accelerator experiments since the 1970s and inserted into textbooks as given — is the output of a quadratic equation whose coefficients are determined by the gauge group and fermion template, both of which were derived from A1.

6.6 The scalar sector

The fermion content has been derived. The hypercharges have been derived. One structural question remains: the fermions are chiral, and chirality (R2) forbids bare gauge-invariant masses. To avoid an infinite admissibility cost at the IR scale, the chiral vacuum must be broken. A scalar field with a vacuum expectation value (VEV) is the minimal mechanism.

Theorem T_{Higgs} (Scalar Sector)

Statement. The admissibility constraints force exactly one complex SU(2) doublet scalar with 4 real components.

Proof. Three steps.
Step 1 (A scalar VEV is required). R2 (chiral carrier) makes the fermion sector chirally asymmetric: left- and right-handed components transform differently under $\mathrm{SU}(2) \times \mathrm{U}(1)$. This forbids gauge-invariant bare mass terms. Without mass generation, the admissibility cost of maintaining massless charged states diverges in the IR: massless gauge-charged particles require long-range correlations, which require unbounded capacity (L_{irr}: costs accumulate irreversibly). An IR-stable vacuum must break the chiral symmetry via a background field with a non-zero VEV [8, 9]. The unbroken EW vacuum ($v = 0$) is therefore inadmissible.
Step 2 (The scalar transforms as an SU(2) *doublet).* To generate gauge-invariant Yukawa couplings to the derived fermion content $\{Q(3,2), L(1,2), u^c(\bar{3},1), d^c(\bar{3},1), e^c(1,1)\}$, the scalar must pair with the SU(2) doublet fermions. The only representation that forms gauge-invariant bilinears with both quark doublets $Q(3,2)$ and lepton doublets $L(1,2)$ is the SU(2) doublet. Singlets and triplets are excluded: a singlet decouples from the doublet fermions (no Yukawa); a triplet generates lepton-number-violating couplings excluded by the derived anomaly cancellation conditions.
Step 3 (One doublet is the minimum by A2). A single complex SU(2) doublet has 4 real components. After spontaneous symmetry breaking $\mathrm{SU}(2) \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\text{em}}$, the 3 Goldstone modes are absorbed by $W^\pm$ and Z (Goldstone theorem [10]: $\dim[\mathrm{SU}(2) \times \mathrm{U}(1)] - \dim[\mathrm{U}(1)_{\text{em}}] = 4 - 1 = 3$), leaving exactly 1 physical massive scalar. A second doublet would add 4 more real components and introduce additional physical scalars and free parameters (two VEVs, a mixing angle, and three extra Higgs bosons) without resolving any structural degeneracy not already resolved by the first doublet. By A2 (minimum cost), the second doublet is not realized: additional admissibility channels without structural necessity have strictly greater total cost than the single-doublet configuration. □

6

The scalar sector contributes 4 structural admissibility channels to the capacity budget. The gauge sector is now complete: gauge group, fermion content, hypercharges, and scalar sector are all derived. Section 7 assembles the total.

[6] Code reference: `check_T_Higgs, check_T_channels` in `gauge.py`.

7. Capacity counting: what constitutes one unit

The previous sections derived the gauge group, the fermion content, and the number of generations. Every piece of the architecture is now in place. This section asks a different question: how large is the budget?

The answer requires a decision that field theory never makes, because field theory never needed to. In quantum field theory, degrees of freedom are counted for the purpose of computing amplitudes and cross-sections. The counting is kinematic: a photon has 2 polarizations, a massive vector boson has 3, a Dirac fermion has 4 spin states. These are the modes that propagate, scatter, and show up in detectors.

But A1 constrains something else. What the interface must hold admissible is not the number of propagation modes but the number of *independent structural distinctions* — equivalently (Paper 1 supplement v8.40 Definition 1 `def:structural-alignment`), the number of independent structural alignments the substrate is committed to hold — the binary questions that the admissibility algebra must resolve to maintain the architecture. A gauge boson has 2 polarizations, but these are propagation modes of a single structural entity: one Lie algebra direction, one admissibility channel. The interface does not need separate capacity to track each polarization; it needs capacity to track whether that gauge direction is *active* at this interface. Similarly, a Weyl fermion has a helicity, but helicity is a kinematic property of a propagating particle, not a structural distinction that the interface enforces. What the interface tracks is whether that chiral species is *present* — a single binary fact.

This distinction between structural and kinematic degrees of freedom has no analogue in standard field theory, where the question never arises. In the admissibility framework, it determines the capacity budget and everything downstream of it.

Before proceeding to the count, it is worth registering a structural fact already established in Paper 1 that shapes everything downstream. T_κ (Directed Enforcement Multiplier) proved that $\kappa = 2$: each independently admissible distinction locks exactly two states. This is not a normalization convention. It means that admissibility is genuinely binary at the base level. Every structural channel answers a single yes-or-no question — is this distinction enforced or not? — and there is no intermediate option. The interface does not set a dial between zero and one; it commits or it does not.

The consequence for the capacity budget is immediate. Each channel contributes an occupation number $n_i \in \{0, 1\}$, not a continuous variable. The 61 structural channels derived below are 61 bits, not 61 continuous degrees of freedom. The admissibility Hamiltonian takes the form $H = -\varepsilon^* \sum_{i=1}^{61} n_i$: a uniform cost per occupied channel, with no cross-coupling terms and no continuous tuning parameters. The ground state of the architecture is a specific bit string — the unique anomaly-free assignment of all 61 channels — not a point in a smooth landscape.

The counting principle rests on a prior lemma establishing what constitutes an independent structural channel.

Lemma L_{species} (Species as Admissibility Channels)

Statement. Each irreducible representation of the derived gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$

that appears in the anomaly-free fermion content defines exactly one independent structural admissibility channel. Kinematic properties of that representation (helicity, polarization, propagation mode) do not contribute additional channels.

Proof. Three steps.
Step 1 (What the interface tracks). By A1, admissibility capacity is finite. What the interface must maintain is not the kinematics of propagating excitations but the *structural distinctions* — the binary facts about which representations are active. For a representation ρ, the binary fact is: is this chiral species present at this interface? That is one admissibility question, costing ε^* per L_{ε^*}.
Step 2 (Kinematic properties are free). Helicity is determined by the Lorentz group representation, not by the gauge admissibility algebra. Once the interface has committed to enforcing that a left-handed electron is present (one capacity unit), its helicity follows from the kinematics of the Lorentz frame (Paper 5); no additional admissibility is required. The same holds for polarizations of gauge bosons: the Lie algebra direction (one structural channel) determines the gauge admissibility role; the two transverse polarizations are kinematic modes of that one channel.
Step 3 (Monogamy enforces independence). By T_M (Monogamy), each distinct irreducible representation requires its own admissibility capacity and cannot share with a different representation. Therefore distinct representations contribute independently, and the count is additive. □ [a]

[a] Code reference: `check_L_count` in `gauge.py`.

The counting principle is then derived from three further ingredients: L_{ε^*} (every distinction has a minimum cost), T_κ (each distinction locks exactly 2 states), and T_M (no two channels share admissibility resources). At Bekenstein saturation [7] — the regime where the interface is maximally packed — every channel costs exactly ε^*, and the total count is simply the number of independent channels.

Lemma L_{count} (Capacity Counting)

Statement. At Bekenstein saturation, the number of independently admissible capacity units equals the number of structural admissibility channels:

$$C_{\text{total}} = \underbrace{45}_{\text{chiral species}} + \underbrace{4}_{\text{Higgs components}} + \underbrace{12}_{\text{gauge directions}} = 61. \tag{15}$$

The count is stated in the channel unit fixed by the cost functional (L_{Cauchy}: $F(n) = n$) under the channel-identification convention — one slot of equal weight per Weyl component, Higgs component, and gauge generator (Supplement bridge axiom BA5).

Proof. Five steps.
Step 1 (L_{ε^*}). Every independently admissible distinction costs $\geq \varepsilon^* > 0$. At saturation, each costs exactly ε^* (maximally packed).
Step 2 (T_κ). $\kappa = 2$: each distinction locks exactly 2 states (binary observable: present vs.

absent).
Step 3 (Structural vs. kinematic DOF).

- **Gauge directions** ($T_3 + T_{\text{gauge}}$): 12 Lie algebra generators. Each is one automorphism direction, regardless of polarization. Polarizations are propagation modes (kinematic), not admissibility structure.
- **Chiral species** (T_{field}): 45 Weyl fermions. Each is one chiral presence. Helicity is kinematic.
- **Higgs components** (T_{Higgs}): 4 real components of the complex SU(2) doublet. Each is one binary VEV-direction distinction.

Step 4 (Independence). By T_M (Monogamy) and L_{loc} (Locality), no two structural channels share admissibility resources. The counting is additive.
Step 5 (Minimality). Each channel resolves exactly 2 states ($\kappa = 2$) and cannot be decomposed further without violating L_{ε^*}. □ [a]

[a] Code reference: `check_L_count` in `gauge.py`.

Why gauge constraints are not subtracted (the BRST question). A field-theorist's instinct is to subtract gauge redundancies via BRST ghost counting, reducing the number of physical degrees of freedom below $\dim(G)$. The admissibility framework does not subtract because the counting principle is different: what A1 constrains is the number of *independently admissible structural distinctions*, not the number of propagating modes. Each Lie-algebra direction is a structural admissibility channel that the interface must independently track, regardless of whether its kinematic content propagates as a physical particle or is removed by gauge-fixing. Technical Supplement I [16], §10 provides the full argument.

7.1 The number

Sixty-one. That is the total admissibility capacity of the universe: 45 chiral fermion species, 12 gauge directions, 4 Higgs components. Every structural distinction that the interface can maintain is accounted for. No parameter has been adjusted. The 45 is the unique minimum among the enumerated candidates of the Section 6 scan. The 12 is $\dim(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)) = 8 + 3 + 1$. The 4 is the real dimensionality of a complex SU(2) doublet. The sum is 61.

This number will determine everything that follows. The partition into vacuum and matter (Section 9) is a partition of these 61 types. The cosmological density fractions are ratios with 61 in the denominator. The number of generations (Section 11) is the largest integer whose capacity cost fits within the slack left by these 61 channels. Sixty-one is not a large number, not a round number, and not a number that anyone would have guessed. It is the number that falls out of the mathematics.

7.2 Cross-checks

Two independent routes yield the same count for the dark sector:

$$\text{Route 1:} \quad 16 = 5 \text{ multiplet types} \times 3 \text{ generations} + 1 \text{ Higgs}, \tag{16}$$

$$\text{Route 2:} \quad 16 = 12\ (\dim G)\ + 4\ (\dim H). \tag{17}$$

This boson-multiplet identity $16 = 16$ is a non-trivial consistency check: two different decompositions of the non-baryonic matter sector agree exactly.

8. The substrate-level reading: Paper 2 derives the content faces of the (61, 102) lattice

The capacity-counting result $C_{\text{total}} = 61$ derived in §7 and the per-channel effective dimension $d_{\text{eff}} = 60 + 42 = 102$ derived in §9.4 below sit inside a substrate-level architecture developed at *Paper 0 v6.1.0 §sec:substrate_position_axis*. The architecture has three regimes (void / middle / saturation) parameterized by where the substrate sits in its capacity budget; four middle-regime commitments characterizing the regime where ordinary physics operates; collapse-as-realignment as the substrate-level reading of the Type III Regime-R exit; and a $(61, 102)$ per-slot channel decomposition giving the channel-level account of which slots fill during a realignment event. Paper 2 derives the *content* that fills the lattice; Paper 0 establishes the substrate-level architecture in which the lattice sits. The two stand in a clean architectural relation, and this short section names that relation explicitly so the reader knows where each load-bearing piece lives.

8.1 Paper 2 in the substrate-position axis

Paper 2 operates inside the substrate's middle regime. A1 sets the upper boundary of admissibility ($C(\Gamma) < \infty$); MD sets the lower boundary ($\varepsilon^*_\Gamma > 0$); together they place the supplement's primitive spine inside the regime where capacity is available, finite, and partly spent — the regime where the four substrate-level middle-regime commitments fire and the substrate is doing its bookkeeping in real time. The *void* regime ($C_\Gamma = 0$ or no admissible-possibility-space data) lies outside Paper 2's scope; the *saturation* regime (where the holographic count ceiling is reached and every admissibility slot is occupied) is reached at the causal horizon and is where L_{equip} (§9) and the proton-stability prediction (§13.3) operate. Paper 2's derivations are middle-regime structural derivations except where Bekenstein saturation is invoked explicitly; the saturation-regime statements are surfaced in their respective subsections.

8.2 The $(61, 102)$ lattice's content faces are Paper 2's territory

The $C_{\text{total}} = 61$ count derived at §7 is the architectural-reading integer of *Paper 0 v6.1.0 §sec:substrate_channel_decomposition*: 61 capacity slots that the substrate must hold admissible at full Bekenstein saturation. The 12 gauge slots (SU(3) at 8, SU(2) at 3, U(1) at 1), the 4 Higgs components, and the 45 fermion slots derived in this paper sit in the *content faces* of the $(61, 102)$ lattice. Each capacity slot has a 60-dim content face carrying its local content and a 42-dim V_{global} vacuum face connecting it to the global vacuum subspace through the Interface-Sector Bridge (Paper 8 supplement v2.18 §1.1 + §H + §E.3). Paper 2 derives what fills the content faces; the substrate-level architecture lives in Paper 0; the bridge mechanism lives in Paper 8 supplement.

Each of the 45 derived fermion fields, in particular, is a *structural alignment* in the sense of Paper 1 supplement v8.40 Definition 1 (`def:structural-alignment`): a configuration the substrate is actually holding, together with the distinctions that configuration carries. Each is held at positive admissibility cost (Paper 1 supplement v8.40 Definition 2, `def:admissibility-substrate`); transitions between them pay the per-realignment floor (Paper 1 supplement v8.40 Definition 3, `def:realignment-cost`). Paper 2 establishes which fermion fields are admissible (1 of 4680 templates survives the anomaly-cancellation filter) and Paper 1 supplement names what each admissible field is in the substrate-level vocabulary.

8.3 Two readings of the partition at two structural depths

The MECE partition $61 = 42 + 19$ derived at §9 admits two readings at two structural depths. The *sector-level* reading is the existing §9 reading: it partitions the 61 capacity slots into 42 vacuum + 19 matter (the matter side further splitting into $3 + 16$ baryonic + dark per the MECE chapter). The *per-slot* reading is what §9.4 derives: each slot's $d_{\text{eff}} = 102$-dim effective space partitions as $60 + 42$, with the 42-dim face being the slot's connection to the global V_{global} subspace via the Interface-Sector Bridge. The two readings are different lattice depths of the same Interface-Sector Bridge content: the sector-level reading partitions the 61-dim representation space; the per-slot reading partitions every slot's 102-element target space. Both readings are required for the framework's substrate-level architecture; *Paper 0 v6.1.0 §sec:substrate_channel_decomposition* develops the per-slot reading at length, and §9.4 below derives the per-slot $60 + 42$ decomposition that anchors it.

8.4 What Paper 2 commits to vs. what Paper 0 commits to

Paper 2 commits to the *content* of the lattice: which gauge group ($\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, derived at §3), which fermions (45 admissible from a 4680-template scan, derived at the fermion-content subsection), which Higgs sector (4 components, derived in the scalar-sector subsection of the gauge-template chapter), what the integer counts are ($C_{\text{total}} = 61$, $d_{\text{eff}} = 102$, $42 + 19$, $42 + 3 + 16$). Paper 0 commits to the *substrate-level architecture* of the lattice: where the substrate sits in its capacity budget (the substrate-position axis), what the four middle-regime commitments are, how realignment events flow capacity through the Interface-Sector Bridge. The two are complementary: Paper 2 cannot derive its content without the substrate-level architecture being in place; Paper 0's architecture is empty without Paper 2's content fillings. The audit-first discipline lives at the boundary — Paper 2 does not claim substrate-level structure beyond what Paper 0 establishes, and Paper 0 does not claim content beyond what Paper 2 derives.

9. The MECE partition: vacuum, baryonic, dark

The previous sections derived the architecture: which gauge group, which fermions, how many generations, how many total capacity types. This section asks how that budget divides up.

The answer is determined by two questions — and only two. Every one of the 61 capacity types either routes through the gauge channels or it does not. Among those that do, each either carries conserved labels protected by confinement or it does not. These two binary predicates generate three sectors, and the proof of exhaustion shows that no third predicate can refine the partition further. The result is a complete accounting of the universe: every capacity type is assigned to exactly one sector, and the fractions are determined by arithmetic.

What makes this partition remarkable is that one of the three sectors — the largest of the matter sectors — is invisible to every particle detector ever built. The framework does not postulate dark matter. It does not add a new field or a new symmetry to accommodate it. Dark matter falls out of the same partition that produces ordinary matter, as the sector that carries gauge-addressable capacity but no conserved color charge. We will unpack what this means after the formal statement.

P_{exhaust} **(Predicate Exhaustion — MECE Partition)**

Predicate Q1 (Gauge addressability, from T_3): Does the capacity unit route through gauge channels $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))$?

- $\mathrm{Q1} = 1 \to$ **matter**: 19 types.
- $\mathrm{Q1} = 0 \to$ **vacuum**: 42 types.

Predicate Q2 (Confinement, within $\mathrm{Q1} = 1$): Does the gauge-addressable unit carry conserved labels protected by SU(3) confinement ($T_{\text{confinement}}$: L_{AF} + A1 saturation $\Rightarrow$ non-singlet states inadmissible)?

- $\mathrm{Q2} = 1 \to$ **baryonic**: 3 types.
- $\mathrm{Q2} = 0 \to$ **dark**: 16 types.

Combined:

$$\underbrace{61}_{C_{\text{total}}} = \underbrace{42}_{\text{vacuum}} + \underbrace{3}_{\text{baryonic}} + \underbrace{16}_{\text{dark}} = \underbrace{42}_{\text{vacuum}} + \underbrace{19}_{\text{matter}} \tag{18}$$

Exhaustion (no third predicate).

- *Vacuum* ($\mathrm{Q1} = 0$): defined by absence of addressable labels. Any splitting predicate would introduce an addressable distinction, moving units to $\mathrm{Q1} = 1$. Contradiction.
- *Dark* ($\mathrm{Q1} = 1$, $\mathrm{Q2} = 0$): gauge-singlet admissibility. Further splitting requires a gauge pathway not in the derived group. T_{gauge} proves $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is complete.
- *Baryonic* ($\mathrm{Q1} = 1$, $\mathrm{Q2} = 1$): indexed by $N_c = 3$, the minimal confining carrier. Sub-ternary splitting would require $\mathrm{SU}(n < 3)$, which does not confine.

[a]

[a]Code reference: `check_P_exhaust` in `core.py`.

The capacity cross-match. A careful reader will notice that the structural count $(45+12+4 = 61)$ and the partition count $(42 + 3 + 16 = 61)$ use different organisational principles. The structural count groups by *field type*; the partition groups by *admissibility mechanism* (predicates Q1 and Q2). The following lemma makes the assignment of each structural class to its sector a proved statement, not a labeling choice.

Lemma L_{vac} **(Vacuum Sector Assignment)**

Statement. Of the 61 structural capacity types, exactly 42 satisfy $\mathrm{Q1} = 0$ (vacuum predicate). Their assignment is determined by three sub-cases, each proved independently.

Case 1: Gauge generators (12 types $\to$ vacuum). Q1 asks whether a capacity type *routes through* the gauge channels. Gauge generators are the gauge channels themselves — they are the admissibility infrastructure, not its consumers. A gauge generator cannot route through itself: it has no "upstream" pathway to enter as a matter excitation. Formally, the generator $T^a \in \mathfrak{g}$ acts as a derivation on the algebra of observables; it is not itself an eigenstate of the Q1 operator. Therefore $\mathrm{Q1} = 0$ for all 12 generators by definition of what "routing through" means.

Case 2: Fermion internal structure (27 of 30 channels → vacuum). The 30 fermion internal-structure channels decompose as: 3 colour charges of the quark sector (Q1 = 1, Q2 = 1, baryonic) plus 27 channels encoding the internal $\mathrm{SU}(N_c) \times \mathrm{SU}(2)$ representation structure of the fermion multiplets. By T_M (Monogamy), each channel has exactly one admissibility role. The admissibility role of the 27 internal-structure channels is to index the representation-theoretic decomposition of the multiplets whose *type identity* is already counted among the 15 dark-sector labels. These 27 are sub-components of that same multiplet role, not independent channels with a separate addressable label. Formally: assigning an independent Q1=1 label to any of the 27 would require either (a) a second admissibility role for the same channel (violating T_M), or (b) a new gauge pathway outside $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ (contradicting T_{gauge}, completeness of the derived group). Neither is possible. Therefore Q1 = 0 for all 27.

Case 3: Higgs components (3 of 4 channels → vacuum). The 4 real Higgs components split as: 1 VEV direction (the physical Higgs after SSB, gauge-addressable, dark sector, Q1 = 1) plus 3 Goldstone directions absorbed by $W^{\pm}$ and Z (Goldstone theorem: $\dim[\mathrm{SU}(2) \times \mathrm{U}(1)] - \dim[\mathrm{U}(1)_{\mathrm{em}}] = 3$ absorbed modes, $\mathrm{T}_{\mathrm{Higgs}}$ Step 3). The 3 absorbed Goldstone modes are longitudinal polarizations of massive gauge bosons — they are part of the gauge infrastructure after SSB, not independently addressable matter labels. Therefore Q1 = 0 for the 3 absorbed components.

Combined count: $12 + 27 + 3 = 42$ vacuum types. □ [a]

[a] Code reference: `check_P_exhaust` in `core.py`.

Table 1 presents the same information in compact form.

Structural origin	**Total**	**Vacuum**	**Baryonic**	**Dark**
Gauge generators ($\mathfrak{su}(3) + \mathfrak{su}(2) + \mathfrak{u}(1)$)	12	12	0	0
Higgs real components	4	3	0	1
Fermion type-identities (5 types × 3 gen.)	15	0	0	15
Fermion internal structure (colour × isospin)	30	27	3	0
Total	**61**	**42**	**3**	**16**

Table 1: Cross-match between the structural count (45+12+4) and the MECE partition (42+3+16), proved in L_{vac}. Gauge generators → vacuum (Case 1 of L_{vac}: they are the admissibility infrastructure, not its consumers). 27 fermion internal channels → vacuum (Case 2: no independent addressable label without duplicating type-identity or extending the gauge group). 3 Higgs components → vacuum (Case 3: absorbed Goldstone modes become longitudinal gauge-boson polarizations after SSB). Remaining 3 baryonic channels are the $N_c = 3$ confined colour charges. 16 dark channels are the 15 multiplet type-identities plus 1 physical Higgs. Verified: $42 + 3 + 16 = 61$; fermion split $27 + 3 + 15 = 45$.

9.1 What dark matter is

For sixty years, dark matter has been the most conspicuous gap in the Standard Model [12]. Galaxies rotate too fast, clusters are too heavy, the cosmic microwave background has the wrong power spectrum — and no particle in the Standard Model can account for it. The standard response has been to postulate new particles: WIMPs, axions, sterile neutrinos, each requiring new fields, new symmetries, and new parameters. Decades of direct detection experiments have found nothing.

The admissibility framework offers a different answer. Dark matter is not a new particle. It is the gauge-singlet sector of the Standard Model's own admissibility structure — the 16 capacity types that are gauge-addressable (Q1 = 1) but carry no conserved color charge (Q2 = 0).

To see what these 16 types are, recall that the matter sector has 19 capacity types total. Three of them carry conserved SU(3) labels protected by confinement: these are the baryonic types, the capacity that tracks the existence of protons and neutrons. The remaining 16 carry gauge quantum numbers — they transform under SU(3) $\times$ SU(2) $\times$ U(1) — but their gauge representation is trivial at the singlet level. They are the admissibility cost of maintaining the Standard Model's matter architecture without producing confined composites.

These 16 types have five properties, each derived rather than assumed. The collisionless and single-fluid properties both rest on the following lemma.

Lemma $L_{\text{singlet,Gram}}$ (Rank-1 Dark Sector)
Statement. The Gram matrix of admissibility demand vectors for the 16 dark-sector capacity types has rank 1.

Proof. Three steps.

Step 1 (Demand vectors for singlets). An admissibility demand vector $\mathbf{v}_i$ for capacity type i encodes how that type loads the local gauge channels. For a type with gauge charges (q_3, q_2, q_1) under SU(3) $\times$ SU(2) $\times$ U(1), the demand vector has components proportional to those charges in the gauge-channel basis.

Step 2 (Dark-sector demand vectors are mutually proportional). The 16 dark-sector types satisfy Q1 = 1 (gauge-addressable) and Q2 = 0 (no conserved SU(3) color charge). Q2 = 0 means the SU(3) components of every dark-sector demand vector are zero. The remaining SU(2) $\times$ U(1) components are determined by the representation assignments derived in T_{field}. Crucially, these 16 types are not gauge singlets—they are matter multiplets (Q, L, u^c, d^c, e^c) with non-trivial SU(2) $\times$ U(1) quantum numbers. What collapses to a single direction is not their charges but their *admissibility demand vectors in gauge-channel space*: since no dark-sector type carries independent SU(3) color charge (Q2 = 0), and since T_M (Monogamy) prohibits each channel from mediating more than one independent admissibility role, the demand vectors of all 16 dark-sector types project onto the same one-dimensional subspace of the gauge-channel vector space—namely, the direction orthogonal to all SU(3) generators. Every dark-sector demand vector is proportional to the identity operator on the SU(3)-neutral subspace—all 16 point in the same direction in that space.

Step 3 (Rank 1). Vectors all proportional to the same direction span a one-dimensional subspace. The Gram matrix $G_{ij} = \mathbf{v}_i \cdot \mathbf{v}_j$ has $G_{ij} = \lambda_i \lambda_j$ for scalar multipliers λ_i, which is an outer product of rank 1. $\square$ [a]

[a]Code reference: `check_L_singlet_Gram` in `cosmology.py`.

- *They gravitate.* All locally committed capacity sources spacetime curvature (T_9: the admissibility cost contributes to the stress-energy tensor). This is not optional — it follows from the same principle that makes ordinary matter gravitate.
- *They are dark.* Their gauge representation is trivial at the singlet level: no electromagnetic coupling, no photon exchange, no interaction with light. Dark matter is dark because the gauge structure derived in Section 3 has no channel through which singlet capacity can emit or absorb photons.
- *They are collisionless at leading order.* The 16 singlet types carry no independent, un-cancelled non-abelian charges (Q2 = 0 by construction). Their admissibility demand vectors therefore have no orthogonal components in the local gauge space; each is strictly proportional to the identity operator on the background volume. Vectors proportional to the same direction span a one-dimensional subspace, so the Gram matrix of the singlet sector has rank 1 ($L_{\text{singlet,Gram}}$). Self-interaction requires at least two linearly independent channels to mediate exchange; with rank 1 there are none: $\sigma/m = 0$ at leading order.
- *They behave as a single fluid.* The rank-1 Gram matrix means the dark sector is one collective mode, not 16 independent species. From the perspective of large-scale dynamics, dark matter is a single substance with a single equation of state. $N_{\text{species}} = 1$, $\Delta N_{\text{eff}} = 0$, consistent with CMB constraints on dark radiation.
- *They are long-lived.* Trivial gauge charge means no decay channel to gauge-charged final states (gauge charge conservation from $L_{\text{anomaly,free}}$). Dark matter does not decay because there is nothing for it to decay into without violating conservation laws that are themselves derived from A1.

The dark matter sector contains 16 of the 19 matter capacity types — it is the dominant component of the matter budget. The baryonic sector, which accounts for all the matter that interacts with light, contains just 3. The ratio $f_b = 3/19$ — three baryonic types out of nineteen matter types — says that ordinary matter is a small minority of the total matter budget. The next subsection derives the bridge that converts these capacity counts into the cosmological density fractions that telescopes measure.

What the framework does *not* claim is equally important. It does not identify a specific particle species for dark matter — dark matter in this framework is not "a particle" but a sector of the admissibility budget. It does not predict small-scale structure (halo profiles, substructure, stream dynamics), which require dynamical evolution beyond the structural level of this paper. And it does not predict sub-leading self-interaction corrections, which depend on higher-order terms in the singlet Gram expansion. These are addressed in Papers 4A and 6.

9.2 From capacity fractions to density fractions

The partition $42 + 3 + 16 = 61$ is an arithmetic fact about capacity types. But cosmologists do not measure capacity types. They measure energy density fractions: the fraction of the universe's total energy budget contained in dark energy, in baryonic matter, in dark matter. The question is whether the capacity fractions *are* the density fractions — and if so, why.

The answer is a short argument with one physical input and one mathematical input. The physical input is that at the causal horizon — the outermost boundary at which admissibility obligations can be coherently maintained — entropy is maximized. This follows from the irreversibility principle (L_{irr}): admissibility commitments accumulate monotonically, and at the horizon, where no further information can arrive, the entropy functional reaches its global maximum (T_{entropy}, Paper 1 [14]).

The argument is microcanonical. At the causal horizon, where L_{irr} drives entropy to its maximum and no further information arrives, the framework can assign no preferred ordering among the 61 capacity types: they are distinguished only by their discrete labels (Q1, Q2 predicates), not by any dynamical weighting. A system with N equally unlabeled microstates has a unique maximum-entropy macrostate: all microstates equally likely. Applied to the 61 discrete types, this gives each type an equal share of the total admissibility capacity.

L_{equip} **(Horizon Equipartition)**

Statement. At Bekenstein saturation (causal horizon), each of the $C_{\mathrm{total}} = 61$ capacity types carries an equal share of the total admissibility capacity: $\varepsilon_i = C/C_{\mathrm{total}}$ for all i. The energy density fraction of any sector equals its type-count fraction:

$$\Omega_{\mathrm{sector}} = \frac{|\mathrm{sector}|}{C_{\mathrm{total}}} \tag{19}$$

Proof. Three steps.
Step 1 (Horizon state). At the causal horizon, $L_{\mathrm{irr}} + T_{\mathrm{entropy}}$ (Paper 1 [14]) drive entropy to its global maximum subject to the constraint $\sum_i \varepsilon_i = C$ and the floor $\varepsilon_i \geq \varepsilon^* > 0$ (L_{ε^*}).
Step 2 (Equal priors from label symmetry). The 61 types are distinguished only by their discrete classification labels (Q1, Q2 predicates applied to the anomaly-free field content). No admissibility mechanism in the framework assigns different dynamical weights to types within the same sector. By the principle of indifference over structurally equivalent microstates, each type receives equal capacity: $\varepsilon_i = C/C_{\mathrm{total}}$. This is the microcanonical result, not a Shannon entropy maximization over probabilities: the argument is that the admissibility ledger has no basis for preferring one type over another within its classification.
Step 3 (Surplus cancellation). The density fraction of sector X is $\Omega_X = \sum_{i \in X} \varepsilon_i \,/\, \sum_{\mathrm{all}\, i} \varepsilon_i = |X| \cdot (C/C_{\mathrm{total}}) \,/\, C = |X|/C_{\mathrm{total}}$. The total capacity C cancels. The fractions depend only on the integer type counts. □ [a]

[a] Code reference: `check_L_equip` in `cosmology.py`.

This is the bridge between structural counting and observational cosmology. It says: if you count the capacity types and sort them into sectors, the fractions you get *are* the density fractions that Planck measures. The bridge requires no dynamical model, no equation of state, no fitting — just constrained entropy maximization over a finite discrete set.

Status of L_{equip}. This lemma is a framework-derived result, not an import from standard cosmology. It does not assume that capacity types are uniformly distributed—it *proves* that they must be, at the causal horizon, by strict concavity of the entropy functional over a finite set with a uniform cost floor (L_{ε^*}). The step that a referee may challenge is the identification of the causal horizon with the cosmological horizon at which Planck's Ω parameters are defined. This identification is established in Paper 6 (*Cosmological Evolution*), which derives the full de Sitter trajectory and confirms that the relevant horizon is the one at which the admissibility ledger reaches Bekenstein saturation. Paper 2 establishes the structural result; Paper 6 closes the cosmological interpretation. At the substrate level: Bekenstein saturation IS the substrate-position-axis saturation regime applied to the $(61, 102)$ lattice. Where L_{equip} holds, every admissibility slot is occupied and each slot carries the same admissibility cost; this is the saturation-regime asymptotic of the substrate's three-regime structure (*Paper 0 v6.1.0 §sec:substrate_position_axis*). The horizon equipartition is therefore not a separate principle imported into Paper 2; it is the substrate-level fact that the framework's middle-regime commitments must reach as the local capacity budget is exhausted.

Under the equipartition bridge's assumption set (bridge axioms BA5–BA7, Technical Supplement I [16]: the equal-unit channel convention, the cosmological identification, and the equipartition conditions), the capacity fractions read as the cosmological density parameters:

Observable	**Capacity fraction**	**Bridge reading**	**Observed**	**Error**
Ω_Λ (dark energy)	42/61	0.6885	0.6889 ± 0.0056 [5]	0.05%
Ω_m (total matter)	19/61	0.3115	0.3111 ± 0.0056 [5]	0.12%
Ω_{DM} (dark matter)	16/61	0.2623	0.2607 ± 0.0056	0.6%
Ω_b (baryonic)	3/61	0.04918	0.0490 ± 0.0003	0.4%
f_b (baryon fraction of matter)	3/19	0.1579	0.1571 ± 0.0012	0.5%

Five bridge readings, zero continuous parameters, all within 0.6% of observation — conditional on BA5–BA7's identifications. The numerators are integers that count capacity types; the denominators are 61. A fit would adjust a continuous parameter to minimize disagreement with data; here, given the identifications, there is no continuous parameter to adjust — the fractions are fixed by the derived field content and the MECE partition. A reader who rejects a bridge axiom loses exactly these readings and nothing upstream (each axiom's failure mode is stated at the ledger); the full comparison and its sensitivity analysis live with Papers 8 and 41.

9.3 Saturation independence

The partition fractions $\Omega_{\text{vac}} = 42/61$ and $\Omega_{\text{mat}} = 19/61$ are *topological invariants* of the matching structure. They depend on discrete type-classification predicates (Q1, Q2), not on the total capacity or saturation level. This topological rigidity is sharpened by the robustness result of the Technical Supplement [15]: if cross-support couplings of size δ perturb exact K3 (disjoint-support additivity), continuous quantities (bilinear form, projections, probabilities) degrade as $O(\delta)$, while discrete invariants—including the Wedderburn block structure, field selection $\mathbb{C}$, and the type-classification predicates that determine the partition—are exactly preserved for $\delta < \delta_{\text{crit}}$. The partition $42 + 3 + 16 = 61$ is therefore structurally stable against perturbations of the cost semantics.

$L_{\text{sat,part}}$ **(Saturation-Independent Partition)**

Statement. The capacity partition $3 + 16 + 42 = 61$ is determined by discrete type-classification predicates applied to the anomaly-free field content. These predicates are functions of type labels, not of total capacity. At any saturation $s > s_{\text{crit}} = 1/d_{\text{eff}} = 1/102$, the max-entropy distribution (L_{equip}) assigns capacity uniformly over all 61 types, and the density fractions are surplus-independent. The threshold $s_{\text{crit}} = 1/102$ marks the substrate's transition from middle-regime non-saturation to the saturation regime where the full matching activates; below the threshold, anomaly cancellation cannot complete because the substrate has not yet committed all 61 channel types simultaneously. The saturation-regime side of this threshold is the asymptotic regime of *Paper 0 v6.1.0 §sec:substrate_position_axis*; below the threshold, Paper 2's middle-regime derivations operate.

$$\Omega_{\text{sector}} = \frac{|\text{sector}|}{C_{\text{total}}} \qquad \text{for all } s > s_{\text{crit}}. \tag{20}$$

Verification. The partition fractions are verified invariant over five decades of surplus $\delta \in \{0, 0.01, 0.5, 5, 100\}$: $\Omega_{\text{vac}} = 42/61$ and $\Omega_{\text{mat}} = 19/61$ at every saturation level. [a]

[a]Code reference: `check_L_saturation_partition` in `cosmology.py`.

Below the critical saturation $s_{\text{crit}} = 1/102$, the full matching does not exist (anomaly cancellation requires all 61 types simultaneously). The pre-matching state is pure de Sitter vacuum with no particle content. The partition is undefined below s_{crit}—but irrelevant, because the vacuum has $w = -1$ regardless.

9.4 Where $d_{\text{eff}} = 102$ comes from

The threshold $s_{\text{crit}} = 1/102$ used above depends on $d_{\text{eff}} = 102$, the effective number of distinguishable states per capacity channel at saturation. This number is not a free parameter; it is determined by the partition just derived.

What states are available to an active channel? At full saturation, each active channel i occupies exactly one of two types of state.

- **Sector A — cross-channel correlations (60 states).** Channel i forms a bipartite correlation with any one of the other $C_{\text{total}} - 1 = 60$ active channels j. This is a genuine admissibility correlation: mutual information $I(i;j) > 0$, cost $\eta(i,j) \geq \varepsilon^* > 0$ (from L_{ε^*}), two distinct endpoints required (T_M: monogamy). The structure is the complete graph K_{61}: every channel has exactly 60 neighbors.
- **Sector B — vacuum background configurations (42 states).** Channel i occupies one of the $C_{\text{vac}} = 42$ gauge-neutral background configurations provided by the vacuum sector. These are not bipartite correlations with specific partners; they are locally entropic states indexed by the 42 vacuum channels. The mutual information is $I(i;j) = 0$: measuring the vacuum label j gives no additional information about i beyond what j's own occupation already tells you. Sector B labels must be gauge-neutral (no conserved flavour quantum numbers) because any conserved quantum number flowing between i and j would constitute a Sector A correlation. The 42 vacuum types (Q1 = 0 from P_{exhaust}) are the unique gauge-neutral index set; the 19 matter types are excluded as Sector B labels.

Why the sectors are disjoint. The two sectors are mutually exclusive states of channel i, not overlapping categories. Sector A requires $I(i;j) > 0$; Sector B requires $I(i;j) = 0$. These are mutually exclusive conditions on the same quantity, enforced by monogamy (T_M): if channel i's admissibility capacity is committed to a bipartite link (Sector A), it cannot simultaneously be in an uncommitted background configuration (Sector B).

Self-exclusion. The raw count before self-exclusion is $d_{\text{raw}} = 60 + 42 + 1 = 103$, where the $+1$ is the candidate self-correlation state. This state is excluded by two independent arguments. (A) *Cost argument*: the self-mutual-information $I(i;i) = H(i)$ is already paid for by channel i's existence cost (L_{ε^*}); the additional cost $\eta(i,i) = I(i;i) - H(i) = 0 < \varepsilon^*$ falls below the minimum admissibility threshold, so the self-correlation is not a meaningful distinction. (B) *Monogamy*: T_M requires two distinct endpoints; the self-correlation (i,i) has only one.

$L_{\text{self-excl}}$**: Effective Dimension per Capacity Channel**

Statement. At saturation, the effective number of distinguishable states per capacity channel is:

$$d_{\text{eff}} \;=\; (C_{\text{total}} - 1) + C_{\text{vac}} \;=\; 60 + 42 \;=\; 102. \tag{21}$$

Inputs: $C_{\text{total}} = 61$ (derived, L_{count}); $C_{\text{vac}} = 42$ (derived, P_{exhaust}); self-exclusion (two proofs above).

Why 102 is an integer. $\kappa = 2$ (T_κ, Paper 1) makes each channel binary: it either enforces its distinction or it does not. Non-integer occupancy would dissolve the channel boundary and render the state count non-integer. The sharpness of $d_{\text{eff}} = 102$—rather than 102-point-something—is a direct consequence of binary admissibility. [a]

[a] Code reference: `check_L_self_exclusion` in `gravity.py`.

The critical saturation threshold now follows immediately: the full matching requires at least one microstate per channel, so $s_{\text{crit}} = 1/d_{\text{eff}} = 1/102$. The full consequences of $d_{\text{eff}} = 102$ for de Sitter entropy and the cosmological constant are developed in Papers 4 and 6.

Remark (why $60 + 42$ and not 60×42). A natural objection: if a channel maintains a bipartite correlation with partner j (Sector A), could it not simultaneously carry a vacuum background label k (Sector B), making the two sectors independent axes with a $60 \times 42 = 2520$-dimensional product?

The answer is no, for two independent reasons.

First, the mutual information criterion is a condition on the *same quantity* $I(i;j)$. Sector A with partner j requires $I(i;j) > 0$; Sector B with vacuum channel k requires $I(i;k) = 0$. These are mutually exclusive values of the same function. For any given partner, the two states cannot coexist.

Second, T_M (monogamy) prohibits simultaneous occupation of both sectors even across different partners. Being in a Sector A state means channel i's capacity is fully committed to maintaining the bipartite correlation with j, at cost $\geq \varepsilon^*$ (L_{ε^*}). Simultaneously occupying a Sector B state with vacuum channel k would require that same capacity budget to be committed to a second admissibility role. Monogamy forbids it: each channel has one admissibility role at a time.

Sector B states are not passive labels. They are active admissibility states that cost ε^* to maintain, in direct competition with Sector A. The microstate space of channel i is the *disjoint union* of the two sectors: $d_{\text{eff}} = 60 + 42 = 102$, not a product. At any moment, channel i is in exactly one state from exactly one sector.

9.5 The vacuum is not empty

The vacuum sector consumes 42 of 61 capacity units—69% of the total budget. This is invisible in the Standard Model because the SM treats the vacuum as given, not as an admissibility cost. In the admissibility framework, the vacuum is the most expensive thing in the universe: maintaining the geometric and gauge structure of empty space costs more than all the matter in it combined.

The identification $\Omega_\Lambda = 42/61 = 0.6885$ (observed: 0.6889 ± 0.0056 [5], error 0.05%) follows from L_{equip} *given the equipartition bridge's assumption set* (equal priors at saturation and the horizon identification; Technical Supplement I, bridge axioms BA5–BA7): at Bekenstein saturation, max-entropy requires each capacity type to carry the same admissibility cost, so the density fraction of any sector equals its type-count fraction. The number 42 counts gauge directions plus Higgs components plus the vacuum-sector fermion types that are globally locked (not locally attributable at any finite interface). The partition is derived, not fitted. The reason the prediction takes the form of an exact fraction — rather than a real number to be fitted or a transcendental constant to be estimated — is T_κ: admissibility is binary, so the partition is a count of bits, and a ratio of two integer bit-counts is necessarily rational. Continuous admissibility would yield continuous density fractions with no reason to be rational; binary admissibility makes rationality inevitable.

Substrate-level reading. The per-channel $d_{\text{eff}} = 60 + 42$ structure derived here is the per-slot face structure of the $(61, 102)$ lattice. Sector A is the slot's content face (60-dim, carrying cross-channel correlations to the other 60 slots); Sector B is the slot's V_{global} vacuum face (42-dim, connecting through the Interface-Sector Bridge to the global vacuum subspace shared across all slots). The sector-level reading of the partition $(42 + 19 = 61)$ and the per-slot reading of d_{eff} $(60 + 42 = 102)$ are two readings of the same Interface-Sector Bridge content at different lattice depths — the sector-level reading partitions the 61-dim representation space; the per-slot reading partitions every slot's 102-element target space. Both readings are needed for the substrate-level account of realignment events; they are developed at *Paper 0 v6.1.0 §sec:substrate_channel_decomposition* and Paper 8 supplement v2.18 §1.1 + §H + §E.3.

10. The Weinberg angle: a preview

The machinery developed in this paper—the unique cost function, the gauge template, the capacity counting—enables the derivation of the weak mixing angle. The full treatment appears in Paper 4A; we state the result here because it is a direct consequence of the structural content of this paper.

The electroweak sector has 4 admissibility channels: 3 mixers (SU(2) generators) plus 1 bookkeeper (U(1) hypercharge). The two gauge sectors compete for capacity at each channel. The Gram matrix of demand vectors is:

$$A = \begin{pmatrix} 1 & x \\ x & x^2+3 \end{pmatrix}, \qquad x = \tfrac{1}{2}, \qquad \det(A) = 3 = \dim \mathfrak{su}(2). \tag{22}$$

The Lyapunov functional governing the coupling flow has a unique UV attractor, with the key input $\gamma_2/\gamma_1 = 17/4$ derived in Paper 4A (T27d) from the cost function uniqueness established in Section 5.

The fixed-point formula gives:

$$\sin^2\theta_W \;=\; \frac{3}{13} \;=\; 0.23077\ldots \tag{23}$$

Observed: 0.23122 ± 0.00004 [4]. Error: 0.19%. All gates closed. [7]

11. Why three generations

Section 6 derived the fermion content per generation—15 Weyl fermions in 5 multiplet types—as the unique minimum among the enumerated candidates. Section 7 established $C_{\text{total}} = 61$. Section 9 partitioned those 61 types into 42 vacuum and 19 matter. One question remains: how many copies of the 15-fermion generation exist?

The answer follows from a capacity ladder, but it is essential to note that the bound is derived *before* counting 61—it depends only on the electroweak sector, which was established in Section 3. This keeps the argument non-circular: the generation bound uses the gauge architecture, and the global count 61 is derived afterwards using the result.

The available capacity for inter-generational tracking is bounded by the electroweak infrastructure capable of routing distinctions between generations. From T_{channels}: the electroweak sector has exactly 4 independent admissibility channels (SU(2): 3 generators; U(1): 1 generator). From T_κ: $\kappa = 2$. The electroweak capacity budget is therefore:

$$C_{\text{EW}} \;=\; \kappa \times \dim_{\text{EW}} \;=\; 2 \times 4 \;=\; 8. \tag{24}$$

This is the capacity available to route inter-generational distinctions. It is derived entirely from Section 3 and does not reference the global count 61.

Each generation costs admissibility capacity in two parts. First, each generation pays a *self-cost*: locking $\kappa = 2$ capacity units for its own existence as a distinct entity with a chirality label ($E_{\text{self}} = \kappa/2 = 1$ in units of ε^*, since $\kappa = 2\varepsilon^*$). Second, each new generation must be distinguished from all previous ones: each pairwise cross-generation distinction costs $E_{\text{pair}} = \varepsilon^* = 1$. With g generations there are $\binom{g}{2} = g(g-1)/2$ such pairs. The total cost is:

$$E(g) \;=\; g \cdot E_{\text{self}} + \binom{g}{2} \cdot E_{\text{pair}} \;=\; g \cdot 1 + \frac{g(g-1)}{2} \cdot 1 \;=\; \frac{g(g+1)}{2}. \tag{25}$$

The quadratic growth is the key: $E(1) = 1$, $E(2) = 3$, $E(3) = 6$, $E(4) = 10$. Each new generation must be distinguished from *all* existing ones, not just the most recent, so the interference cost grows faster than linearly.

Now evaluate the ladder against $C_{\text{EW}} = 8$:

[7] Code reference: `check_L_W_mass, check_L_dim_angle` in `gauge.py`.

g	$E(g) = g(g+1)/2$	$C_{\text{EW}} = 8$	Admissible?
1	1	8	Yes
2	3	8	Yes
3	6	8	Yes
4	10	8	**No**

Three generations fit within the electroweak capacity budget. Four do not. $E(3) = 6 \leq 8 < 10 = E(4)$. The quadratic growth of the interference cost—the same superadditivity mechanism that drove non-closure in Section 2—is what makes the wall sharp.

This is the same logic that excluded non-singlet states in Section 3: a combination of individually affordable commitments becomes collectively unaffordable because of the interference surplus. The first three generations coexist because their mutual interference costs fit within the electroweak budget. A fourth generation would push the total past the capacity wall.

The number of generations is not a free parameter. It is the largest integer g satisfying $E(g) \leq C_{\text{EW}}$. The inputs—κ, ε^*, and $C_{\text{EW}} = \kappa \times \dim_{\text{EW}}$—are all derived from A1 and the gauge architecture of Section 3. The global count of 61 plays no role in this derivation; the generation bound is established before the full budget is assembled.

Flavor-capacity saturation. The three-generation cost $E(3) = 6$ consumes exactly $6/8 = 3/4$ of the electroweak capacity budget (T_{4F}, gauge.py). This near-saturation has a structural consequence: the remaining 2 units of electroweak capacity are insufficient to distinguish a fourth generation but sufficient to support the cross-generation mixing encoded in the CKM and PMNS matrices. The mass hierarchy across generations—$m_t \gg m_c \gg m_u$—follows from capacity ordering: the generation that costs the least admissibility overhead to maintain acquires the largest share of the Yukawa budget (T_{4E}). The quantitative derivation of mass ratios and mixing angles is deferred to Paper 4.

κ-class invariance. The qualitative algebraic skeleton (Hilbert space, C^*-algebra, Born rule, Tsirelson bound) is structurally invariant across the entire admissible κ-class: any cost function satisfying K1–K3 produces the same inter-sector orthogonality, the same projection structure, and the same downstream algebra (Proposition κ_{class}, Technical Supplement [15]). The value $\kappa = 2$ is the specific derived value for the physical interface (T$_\kappa$, Paper 1); the generation bound $N_{\text{gen}} = 3$ uses this value quantitatively and would change if κ differed, but the gauge architecture and noncommutativity that produce the generation ladder are κ-independent. [8]

12. The hydrogen atom revisited

The introduction opened with a hydrogen atom drifting in deep space and used it to motivate non-closure before any machinery was in place. The machinery is now in place. We can say precisely what the hydrogen atom is in the admissibility framework—and precisely what we cannot yet say.

The count is over-determined, in both directions. The ladder argument gives the ceiling. The floor — why the multiplicity is not one or two — does not rest on the ladder alone. Three independent capacity–beta identities each solve uniquely to $n = 3$: $6|b_3| = C_{\text{vacuum}}$, $6|b_2| = C_{\text{matter}}$[9],

[8]Code reference: `check_T7, check_T4E, check_T4F` in `gauge.py`.
[9]Code reference: `check_L_beta_capacity` in `generations.py`.

and $6|b_Y| = d_{\text{eff}} - C_{\text{total}}$[10]. These are equations, not bounds: $n = 1, 2$, and 4 fail all three. Under-replicated rivals are additionally killed by the max-saturated selection[11]. The practical consequence for the fermion-content scan of Section 6 is worth stating plainly: every candidate content is evaluated at the forced multiplicity, so a rival that is admissible only as a single copy — the colored weak-triplet content of the supplement's Example 4 is the sharpest case — is excluded before the content filters fire.

The channels present

The electron is a Gen 1 lepton. Its type-identity is carried by three channels in the 61:

- L_1, L_2 — the lepton doublet, gauge-addressable under SU(2), dark sector (Q1 = 1, Q2 = 0).
- e_c — the right-handed electron singlet, gauge-addressable under U(1), dark sector.

The electroweak gauge channels active at this energy are W_1, W_2, W_3 (SU(2), dark sector) and B (U(1), dark sector). The photon is the post-symmetry-breaking mixture of B and W_3; the Coulomb binding of the electron to the proton routes through these two channels. That is 7 channels from the 61, all in the dark sector (Q1 = 1, Q2 = 0).

Three classes of channels are *absent*, and their absence is a structural prediction:

- SU(3) **channels** $(g_1, \ldots, g_8)$: not involved. The electron is colorless; no SU(3) admissibility is required.
- **Higgs channels** $(H_r^{\pm}, H_i^{\pm}, H_r^0, H_i^0)$: not active at hydrogen energies.
- **Gen 2 and Gen 3 channels**: not involved. Generation identity is a quantum-number value, not a mechanism-level channel distinction. The electron does not draw on muon or tau channels.

The shared pool

Both spin admissibility and position admissibility draw additionally from a subset of the 42 vacuum-sector channels at the electron's locus—the channels maintaining the local geometric structure. Spin is a representation of the local Lorentz group $SO(1, 3)$; in the framework, the Lorentz frame at a locus is maintained by a subset of the vacuum-sector channels derived in Paper 5 from the vielbein and spin connection. Position is a distribution of the electron's type channels across multiple spatial loci; the distances between those loci are maintained by a further subset of the vacuum-sector channels, derived from the cost metric (L_{cost}) and the polarization identity (T_{7B}), also in Paper 5.

The interference is real and structural: spin and position both draw from the same local vacuum-sector budget, their joint admissibility cost exceeds the sum of their individual costs by the surplus $\Delta > 0$, and that surplus is the admissibility-language statement of the Heisenberg uncertainty relation $\Delta x\, \Delta p \geq \hbar/2$.

What this paper cannot yet say

This paper cannot name the specific vacuum-sector channels that carry Lorentz frame admissibility or spatial metric admissibility. The identification requires the spacetime derivation of Paper 5. What this paper *can* say is:

[10] Code reference: `check_T_gauge_beta_capacity_tiling_abelian` in `gauge_beta_capacity_tiling.py`.
[11] Code reference: `check_R_Ngen_neq_3_killed` in `killed_rivals.py`.

1. The channels exist and are vacuum-sector (Q1 = 0), drawn from the 42.
2. They constitute the local geometric infrastructure that both spin and position tasks must share.
3. The interference $\Delta > 0$ is a consequence of that sharing, not of scarcity.
4. The Heisenberg bound is the quantitative expression of non-closure at this interface.

The hydrogen atom opened this paper as an intuition. Paper 5 closes it as a derived result: the same example, fully accounted for at the channel level, with every vacuum-sector contribution named.

The proton: non-closure as exclusion

The proton consists of two up quarks and one down quark. Its type-identity draws on 12 fermion channels from the 61, all Gen 1, all baryonic sector (Q1 = 1, Q2 = 1):

- $Q_1, \ldots, Q_6$ — the quark doublet $Q(\mathbf{3}, \mathbf{2})_{1/6}$: 6 channels (3 colours $\times$ 2 SU(2) components).
- u_1^c, u_2^c, u_3^c — right-handed up singlet: 3 channels.
- d_1^c, d_2^c, d_3^c — right-handed down singlet: 3 channels.

The gauge channels active at hadronic energies are all 8 gluon channels $g_1, \ldots, g_8$ (SU(3), vacuum sector), together with W_1, W_2, W_3 (SU(2)) and B (U(1)). That is 24 channels from the 61 in total—more than three times the 7 active at the electron.

Three classes of channels are again absent:

- **Lepton channels** (L_1, L_2, e_c, and their generational copies): not involved. The quarks carry colour; the leptons do not.
- **Higgs channels**: not active at these energies.
- **Gen 2 and Gen 3 channels**: not involved for a ground-state proton.

The mechanism of confinement is not an additional postulate. It is $T_{\text{confinement}}$ (§3): asymptotic freedom (L_{AF}) means the SU(3) coupling grows as the interface scale increases toward the infrared. The admissibility cost of maintaining a non-singlet coloured state grows with it. At the confinement scale, that cost exceeds the available local capacity—the non-singlet state overflows the budget and is excluded. The surviving configuration is the colour-singlet proton: three quarks whose SU(3) charges cancel exactly, costing nothing to the SU(3) budget at the interface.

The electron showed non-closure as interference: both tasks survive, the joint cost is elevated, Heisenberg is the result. The proton shows non-closure as exclusion: the coloured state does not merely cost more, it cannot exist. Same mechanism, different severity, determined entirely by the structure of the SU(3) channels and the capacity budget.

What this paper cannot say about the proton is the *scale* at which exclusion occurs—the value of Λ_{QCD}, the string tension, the hadron spectrum. Those require quantitative beta coefficients derived in Paper 4A. The mechanism is derived here; the scale is not.

The hydrogen atom: a two-body system

The hydrogen atom as a bound state draws simultaneously on both channel sets. The electron contributes L_1, L_2, e_c (matter channels) and shares W_1, W_2, W_3, B (gauge channels) with the proton. The proton contributes $Q_1, \ldots, Q_6$, u_1^c, u_2^c, u_3^c, d_1^c, d_2^c, d_3^c (matter channels) and $g_1, \ldots, g_8$ (gluon channels, absent at the electron but present at the proton's locus). The photon—the B-W_3 mixture—mediates the Coulomb binding between the two loci.

The total channel count active in a hydrogen atom: 24 from the proton side, 3 additional lepton channels from the electron side, the shared 4 electroweak channels, plus vacuum-sector geometric channels at each locus. The SU(3) channels are active at the proton's locus and screened at the electron's locus—the electron does not see the colour field because it carries no colour charge.

Channel	**Sector**	**Electron locus**	**Proton locus**
$g_1, \ldots, g_8$	vacuum (gauge)	absent	active
W_1, W_2, W_3	dark	active	active
B	dark	active	active
$Q_1, \ldots, Q_6$, u_{1-3}^c, d_{1-3}^c	baryonic	absent	active
L_1, L_2, e_c	dark	active	absent
vacuum geometric	vacuum	active	active
Total (named)		**7**	**24**

Table 2: Channel activity in hydrogen by locus. Vacuum-sector geometric channels (Lorentz frame, spatial metric) are present at both loci but not individually named until Paper 5.

The hydrogen atom is not exotic physics. It is the simplest system in which all three faces of non-closure are simultaneously present: interference (electron spin and position), exclusion (quark confinement inside the proton), and cost minimisation (the bound state is the cheapest combination that survives both constraints). The framework derives all three from A1. Paper 5 will name every channel in the table.

13. Three structural predictions

The machinery of this paper—capacity counting, the MECE partition, and the unique cost function—yields three predictions that go beyond the Standard Model's architecture and address open questions that standard field theory leaves unanswered.

13.1 The strong CP problem is dissolved: $\theta_{\text{QCD}} = 0$

The QCD vacuum angle θ parameterises a topological term $\mathcal{L}_\theta = (\theta/32\pi^2)\, G\tilde{G}$ that is a total divergence. It introduces no new propagating degrees of freedom and leaves $C_{\text{total}} = 61$ unchanged. However, maintaining a definite value $\theta = \theta_0 \neq 0$ is an admissible distinction: distinguishing θ_0 from other values in $[0, 2\pi)$ costs at least $\varepsilon^* > 0$ by L_{ε^*}. The value $\theta = 0$ is unique: it is the only value at which the strong sector is CP-invariant, so $\theta = 0$ is not an independent parameter but a consequence of the symmetry. No admissibility record is needed to maintain a symmetry consequence; the cost is zero. Both $\theta = 0$ and $\theta \neq 0$ are admissible under A1: each respects $\Sigma\varepsilon \leq C$. But $\theta = 0$ is admissible at zero maintenance cost (the symmetry consequence carries no admissibility record), while $\theta \neq 0$ is admissible only at $\varepsilon^* > 0$. Since the two configurations realise the same structural content but at strictly different total cost, A2 (minimum cost) selects $\theta_{\mathrm{QCD}} = 0$ as the realised value over the admissible set.

The result dissolves the strong CP problem: there is no parameter to tune and no axion to postulate. The experimental bound $|\theta| < 10^{-10}$ [4] is satisfied exactly, not approximately. [12]

13.2 The electroweak vacuum is absolutely stable

In the Standard Model, the Higgs effective potential extrapolated to high energies via renormalisation group running develops a second minimum deeper than the electroweak vacuum for $m_H \approx 125$ GeV and $m_{\mathrm{top}} \approx 173$ GeV, rendering the vacuum metastable. In the admissibility framework, this second minimum does not exist.

The admissibility potential $V(\Phi)$ derived from L_{ε^*}, T_M (monogamy binding), and A1 (capacity saturation) has a unique global minimum: $V(0) = 0$ (empty vacuum, unstable), a barrier at $\Phi/C \sim 0.06$, a unique binding well at $\Phi/C \sim 0.73$ with $V < 0$, and a divergence at $\Phi \to C$ (capacity saturation). The divergence at $\Phi \to C$ prevents any second minimum from forming at high field values. The electroweak vacuum occupies the unique well and is absolutely stable—not metastable with a long lifetime, but stable as a matter of admissibility structure. [13]

The hydrogen atom as a worked example of the four middle-regime commitments. The channel-level account developed in this section is the framework's worked example of the four substrate-level middle-regime commitments named at *Paper 0 v6.1.0 §sec:substrate_middle_commitments*. The electron + proton + photon system is a structural alignment held by the substrate at positive admissibility cost (commitment C1 — the electron's 7 channels and the proton's 24 channels are real because the substrate has paid to hold them); the realignments between admissible configurations (orbital transitions, photon emission/absorption) pay the per-realignment floor (commitments C2 and C3); the spin-and-position joint correlation costs interface capacity (commitment C4 — the source of the interference surplus Δ developed at §1); the system can move from one admissible configuration to another because admissible destinations exist in the middle regime. The hydrogen atom is the four middle-regime commitments at one worked physical scale; §8 above locates the chapter's content derivation in the substrate-level architecture.

13.3 The proton is absolutely stable

At full Bekenstein saturation, the three-sector partition $N_b + N_d + N_v = 3 + 16 + 42 = 61 = C_{\mathrm{total}}$ is exact (P_{exhaust}): every capacity type is assigned to exactly one sector, with no ambiguity. The baryonic stratum has $N_b = 3$ types, and this integer is a conserved quantity at saturation. At the substrate level, this is a saturation-regime structural prediction: the exactness of the partition AT Bekenstein saturation is what locks the baryonic count, and the saturation-regime asymptotic of the $(61, 102)$ lattice (*Paper 0 v6.1.0 §sec:substrate_position_axis*; Paper 36 v0.3 *The Missing Floor* for the holographic count ceiling reading) is what makes the prediction structural rather than approximate. Proton stability is therefore Paper 2's saturation-regime contribution to the framework's substrate-position-axis predictions, sitting alongside the cosmological density fractions

[12] Code reference: `check_T_theta_QCD, check_L_strong_CP_synthesis` in `gauge.py`.
[13] Code reference: `check_T_vacuum_stability` in `gauge.py`.

and the gauge-content count as predictions whose substrate-level reading lives in the saturation regime. Any process converting a baryon-sector type to a non-baryon type would violate the MECE partition, which is a mathematical identity at capacity saturation, not an approximate symmetry.

Proton decay would require a capacity transfer from the baryonic sector ($N_b = 3$) to the leptonic or gauge sector, changing N_b. At saturation, this is forbidden: the partition is sharp and the sector counts are fixed. The proton is therefore absolutely stable within the framework—not stable with a lifetime exceeding experimental bounds, but stable as a consequence of the exact partition. One clause of earlier versions is corrected here: gauge invariance does *not* exclude baryon-number-violating operators as constructions — the $\Delta B = 1$ dimension-six operators of the SMEFT basis are gauge-invariant under $SU(3) \times SU(2) \times U(1)$. What the framework establishes is that nothing in the *derived particle content* generates them: no mediator with leptoquark or X/Y quantum numbers exists, and the exclusion of additional states closes the loop route. The $\tau_p = \infty$ conclusion is therefore conditional on the completeness of the derived content; ultraviolet degrees of freedom outside it could source these operators while respecting the gauge group. [14]

14. The correlation space after this paper

Paper 1 [14] established that the correlation space $\mathcal{S}_{\text{adm}}$ is a finite-dimensional convex body inside the space of density operators, carrying an operator algebra, a gauge symmetry, and an entropy functional. We knew its mathematical type but not its size, its symmetry group, or its physical content.

After Paper 2, we know:

1. *It has rooms.* The 61 capacity types define the dimensionality of the admissibility structure. The correlation space is stratified by the partition $42 + 19 = 61$ into a 42-dimensional vacuum chamber and a 19-dimensional matter chamber.

2. *The floor plan is unique.* The gauge group $SU(3) \times SU(2) \times U(1)$, the fermion content (45 Weyl fermions in 3 generations), and the Higgs sector (4 real components) are the unique architecture consistent with A1. No alternative exists.

3. *The partition is rigid.* The fractions $\Omega_{\text{vac}} = 42/61$ and $\Omega_{\text{mat}} = 19/61$ are topological invariants of the matching structure. They cannot evolve, cannot be tuned, and cannot depend on continuous parameters.

4. *Competition is quantified.* Non-closure (L_{nc}) provides the mechanism, the unique cost function (L_{Cauchy}) provides the metric, and the capacity count (L_{count}) provides the budget. The framework can now compute which combinations survive and which are excluded.

The correlation space now has structure. It has a gauge architecture, a particle content, and a budget partition. What it does not yet have is a direction—an arrow of time, a thermodynamic interpretation, an entropy gradient. That is Paper 3: *Ledgers*.

[14] Code reference: `check_T_proton, check_L_B_operator_exclusion` in `gauge.py`.

15. What this paper does not claim

Several important results are *previewed* in this paper but *derived* in later papers. To maintain logical clarity:

- **The number of generations** ($N_{\text{gen}} = 3$) is derived in Section 11 from the capacity ladder: the quadratic growth of inter-generation interference costs makes four generations inadmissible. The inputs (κ, ε^*, $C_{\text{EW}} = 8$) are all derived from A1 and the gauge architecture.
- **The Weinberg angle** is previewed in Section 10 but fully derived in Paper 4A. This paper establishes the cost function uniqueness that the derivation requires.
- **Cosmological density fractions** ($\Omega_\Lambda = 42/61$, $\Omega_m = 19/61$) are derived in Section 9 from the capacity partition and horizon equipartition (L_{equip}). The *dynamical interpretation*—what these fractions mean for the expansion history, the cosmological constant magnitude ($\Lambda \cdot G = 3\pi/102^{61}$), and confrontation with CMB anisotropy data—is the subject of Paper 6.
- **Confinement mechanism class** is derived in this paper ($T_{\text{confinement}}$, Section 3): asymptotic freedom (L_{AF}) plus IR saturation (A1) makes non-singlet states inadmissible. The β-function sign is verified with the imported coefficient in Technical Supplement I [16], §2 (capacity-side heuristic: Technical Supplement II [17], §4). What remains open—and is addressed in Paper 4A—is the confinement *scale* (Λ_{QCD}), string tension, and hadronization dynamics, which require quantitative beta coefficients.
- **Mass hierarchy, mixing matrices, and CP violation** are derived in Papers 4A and 4B. This paper establishes the gauge and matter architecture; the quantitative predictions require the generation structure.
- **Spacetime derivation.** The chain to Lorentzian signature (Δ_{sig}), spacetime dimension $d = 4$ (T_8), and the Einstein field equations ($T_{9,\text{grav}}$) lives at Paper 6; Technical Supplement I [16] carries the exit map and the bridge-axiom ledger (BA1–BA4) naming each imported hypothesis and its failure mode.
- **Neutrino masses.** The scan fact is sharp: the minimum template contains no light sterile singlet (Technical Supplement I [16], §9). The ν_R-extended template (48 Weyl fermions) moves the count to $C_{\text{total}} = 64$, whose bridge reading $\Omega_\Lambda = 42/64 = 0.656$ (BA6) is excluded at $> 5\sigma$ by Planck [5] — a bridge-conditional comparison that consumes BA6 and carries its failure mode. The Majorana/$0\nu\beta\beta$ prediction register and the full bridge analysis live at the cosmology-tail corpus homes (Papers 8 and 41).

This paper makes *no* claim that could not be stated at the structural level. It counts rooms, draws the floor plan, and establishes the budget. What happens inside the rooms—the dynamics, the masses, the mixing—is the subject of the remaining papers.

16. How to falsify this paper

The structural claims of this paper are sharp enough to be falsified. Each corresponds to a specific challenge:

- **(F1) An A1-compatible theory without R1–R3.** Construct a consistent interaction theory obeying finite admissibility capacity that does not require a ternary carrier, a chiral carrier, or an abelian grading. This would falsify Stage 1 of the three-stage argument.

- **(F2) A smaller gauge group supporting R1–R3.** Exhibit a compact gauge group with $\dim(G) < 12$ that hosts all three carrier requirements. This would falsify Stage 2.
- **(F3) An alternative cost functional.** Construct a cost functional compatible with A1 (additive, monotone, positively normalized) that is not $C(G) = \dim(G) \cdot \varepsilon$. This would falsify Stage 3 and potentially select a different minimal algebra.
- **(F4) Alternative fermion content.** Exhibit a chiral fermion set with fewer than 45 Weyl degrees of freedom that satisfies all seven filters (asymptotic freedom, anomaly cancellation, doublet–singlet content, Witten freedom, minimality). This would falsify T_{field}.
- **(F5) Alternative capacity counting.** Demonstrate that kinematic degrees of freedom (polarizations, helicities) are structurally admissible at Bekenstein saturation. This would change C_{total} from 61 to 73 (or another value) and falsify the MECE partition fractions.

Experimentally, the following observations would falsify specific results of this paper:

- **(F6)** A fourth fermion generation participating coherently without saturation-like behavior (falsifies $N_{\text{gen}} = 3$).
- **(F7)** Dirac neutrinos with standard electroweak couplings (changes C_{total} from 61 to 64, whose bridge reading $\Omega_\Lambda = 42/64 = 0.656$ (BA6) is excluded at $> 5\sigma$ by Planck [5]). The 45-fermion vs. 48-fermion discrimination is the sharpest single-observable test of T_{field}; the scan fact is Technical Supplement I [16] §9, and the bridge-conditional analysis lives at the cosmology-tail corpus homes (Papers 8 and 41).
- **(F8)** Dark matter detected as a fundamental particle species not in the Standard Model field content (falsifies T_{field} completeness).

Constructive counter-models are welcomed.

17. Summary

This paper has deployed the results of Paper 1 [14] and established the structural architecture of admissible physics:

Result	**Theorem**	**Key content**	**Code reference**
Non-closure engine	L_{nc}	$\Delta > 0$ derives competition	`check_L_nc`
Confinement	T_{conf}	IR saturation $\to$ singlets	`check_T_confinement`
Cost uniqueness	L_{Cauchy}	$F(d) = d$ from Cauchy 1821	`check_L_Cauchy_uniqueness`
Gauge template	$L_{\text{gauge,uniq}}$	families + exceptionals $\to$ unique	`check_L_gauge_template_uniqueness`
Gauge group	T_{gauge}	$\text{SU}(3)\times\text{SU}(2)\times\text{U}(1)$	`check_T_gauge`
Fermion content	T_{field}	1,680 $\to$ 1 survivor	`check_T_field`
Capacity count	L_{count}	$45 + 4 + 12 = 61$	`check_L_count`
MECE partition	P_{exhaust}	$42 + 3 + 16 = 61$	`check_P_exhaust`
Saturation indep.	$L_{\text{sat,part}}$	Fractions are topological	`check_L_saturation_partition`
Eff. dimension	$L_{\text{self-excl}}$	$d_{\text{eff}} = 60 + 42 = 102$	`check_L_self_exclusion`
Generations	T_7	$N_{\text{gen}} = 3$ from capacity ladder	`check_T7`
Strong CP	$T_{\theta_{\text{QCD}}}$	$\theta = 0$ from minimality	`check_T_theta_QCD`
Vacuum stability	$T_{\text{vac,stab}}$	Unique admissibility well	`check_T_vacuum_stability`
Proton stability	T_{proton}	Exact B conservation at saturation	`check_T_proton`

Continuous free parameters introduced: 0; the discrete premises of the classification — the scan caps, F6's quark-doublet hypercharge non-degeneracy, generation universality, the admissibility-completeness reading, and the equal-unit channel convention — are named at their points of use. The Standard Model's gauge group and matter content are derived from a capacity optimization problem stated in the language of A1 (with MD pricing the floor and A2 selecting the argmin); the three structural predictions carry their own grades — strong CP at [P-heuristic], vacuum stability within the framework's own potential, and proton stability conditional on the completeness of the derived content.

Paper 2's $K = 61$ at the T_ACC seam (v5.4). As of the current codebase (v24.3.423), the integer $C_{\text{total}} = 61$ derived above in §"Capacity counting: what constitutes one unit" (L_{count}) is the structural-enumeration reading of a single capacity number that reappears at the cosmological-partition seam of Paper 6. T_ACC identity $I2$ (gauge-cosmological) states the seam formally: the integer K_{SM} that the gauge-regime projection π_F reads at the Standard-Model interface equals the denominator that the cosmological-regime projection π_C writes for every Friedmann fraction: $\Omega_b = 3/61$, $\Omega_c = 16/61$, $\Omega_\Lambda = 42/61$ [15]. Paper 2 establishes the integer and the partition; Paper 3 Supplement §7 formalizes the ledger; Paper 6 Supplement §sec:Bridge_compressed extends the identity to the 42-dim subspace V_{global} (strictly finer than $I2$). The 1.2% match between APF's {3/61, 16/61, 42/61} and Planck 2018 is corroborating evidence for the seam; the seam itself is an integer-level structural claim. See Technical Supplement I §"$I2$ (gauge-cosmological identity): the $K = 61$ shared integer" (inserted after the L_{count} subsection) for the formal statement and the interactive figure reference (`Fig_61_Channels_Bridge_to_102_v6.html` in the Paper 0 folder).

[15] Code reference: `check_I2_gauge_cosmological` in `unification.py`.

18. Appendix A: theorem index

Name	**Full title**	**Dependencies**	**Code / P2S**
L_{nc}	Non-Closure	A1, L_{loc}; P1S: L_{Δ}	`check_L_nc`
L_{Cauchy}	Cost Function Uniqueness	A1, L_{loc}, L_{cost}	`check_L_Cauchy_uniqueness`; P2S §1
T_{7B}	Cost Kernel (pos. def.)	A1, K2, K3, SP, L_{ω}, T_{adj}	P2S §3.1
$L_{\text{irr,unif}}$	Sector-Uniform Irrev.	L_{loc}, L_{nc}, L_{irr}, T_{7B}	P2S §3.2
L_{AF}	Asymptotic Freedom	T_3, $\dim(\text{adj}) > 0$	`check_T_confinement`; P2S §4
T_{conf}	Confinement Mechanism	L_{AF}, A1, L_{ε^*}	`check_T_confinement`; P2S §4
B_1'	Complex Carrier	T_{conf}, T_M, rep. theory	`check_B1_prime`; P2S §3.3
Theorem R	Repr. Requirements	L_{nc}, $L_{\text{irr,unif}}$, T_M, T_{conf}, B_1'	`check_Theorem_R`
$L_{\text{gauge,uniq}}$	Gauge Template Uniq.	Thm R, L_{Cauchy}, Lie classif.	`check_L_gauge_template_uniqueness`; P2S §II.1
T_{gauge}	Gauge Group Selection	T_4, T_5, L_{cost}	`check_T_gauge`
T_{field}	Fermion Content	T_{gauge}, T_7, L_{AF}	`check_T_field` [18]; P2S §II.2
$L_{\text{hypercharge}}$	Hypercharge Uniq.	T_{field}, anomaly conditions	P2S §II.3
L_{species}	Species = Channels	T_M, L_{ε^*}, T_{field}	(within `check_L_count`); P2S §III.1
L_{count}	Capacity Counting	L_{species}, T_{κ}, T_3	`check_L_count`; P2S §III.1
P_{exhaust}	MECE Partition	A1, T_3, T_{gauge}, T_4	`check_P_exhaust`; P2S §III.2
L_{equip}	Horizon Equipartition	P_{exhaust}, L_{irr}, L_{ε^*}	`check_L_equip`; P2S §III.3
$L_{\text{sat,part}}$	Saturation Indep.	L_{equip}, L_{count}; P1S: δ_{crit}	`check_L_saturation_partition`; P2S §III.4
$L_{\text{self-excl}}$	Effective Dimension	L_{ε^*}, T_M, T_{κ}, L_{count}, P_{exhaust}	`check_L_self_exclusion`
T_{channels}	EW Channels = 4	T_{gauge}, $\dim \mathfrak{su}(2) + 1$	`check_T_channels`; corollary of P2S §II.1
T_7	Generation Bound $N_{\text{gen}} = 3$	T_{κ}, T_{channels}, L_{ε^*}	`check_T7`; §11
T_{4E}	Generation Structure	T_7, T_{4F}, capacity ordering	`check_T4E`
T_{4F}	Flavor-Capacity Saturation	T_7, $C_{\text{EW}} = 8$	`check_T4F`
$T_{\theta_{\text{QCD}}}$	Strong CP: $\theta = 0$	L_{ε^*}, A2	`check_T_theta_QCD`; §13.1
$T_{\text{vac,stab}}$	Vacuum Absolute Stability	L_{ε^*}, T_M, A1 (unique well)	`check_T_vacuum_stability`; §13.2
T_{proton}	Proton Absolute Stability	P_{exhaust}, saturation exactness	`check_T_proton`; §13.3

All check functions reside in `core.py`, `gauge.py`, `cosmology.py`, `gravity.py`, or `L_Cauchy_uniqueness.py`. The fermion-content derivation (T_{field}) additionally has a standalone verification script `fermion_scan_standalone.py` [18] with an interactive Colab walkthrough. "P1S" denotes the Paper 1 Technical Supplement [15]; "P2S-I" denotes Paper 2 Technical Supplement I (*The Classification Core*) [16] and "P2S-II" Technical Supplement II (*The Foundational Gauge Program*) [17].

19. Appendix B: symbol index

Symbol	**Meaning**	**Defined**
From Paper 1 [14] and Technical Supplement [15] (used throughout)		
A1	Finite admissibility capacity (PLEC upper bound on admissibility space)	P1, §2
A2	arg min selection (PLEC variational component)	P1, §2
L_{nc}	Non-Closure lemma	P1, §4.3
L_{irred}	Irreducibility of the derived algebra	§3
L_{loc}	Locality lemma	P1, §4.2; P1S, T_{loc}
L_{irr}	Irreversibility lemma	P1, §4.5
L_{Δ}	Superadditivity ($\Delta > 0$ at shared interfaces)	P1S, §3
L_{Π}	Codespace rotation (joint defender rotated into pool)	P1S, §4
T_{alg}	Noncommutativity of admissibility algebra	P1S, §4
T_{sector}	Finite orthogonal sector form	P1S, §3
T_M	Interface monogamy	P1, §5.10; P1S, T_M
T_3	Gauge structure (Skolem–Noether)	P1, §5.7
ε^*	Minimum admissibility cost	P1, §4.1; P1S, L_{ε^*}
κ	Directed multiplier ($= 2$)	P1, §5.11
Introduced in this paper (proofs: Technical Supplements I and II [16, 17])		
L_{AF}	Asymptotic freedom ($\dim(\mathrm{adj}) > 0 \Rightarrow$ UV fixed point)	§3; P2S §4
$T_{\mathrm{confinement}}$	Confinement mechanism (IR saturation $\to$ singlets only)	§3; P2S §4
B_1'	Complex carrier ($V \not\cong V^*$ derived)	§3; P2S §3.3
T_{7B}	Positive-definite cost kernel at single interface	P2S §3.1
$L_{\mathrm{irr,uniform}}$	Gauge-sector irreversibility at saturated interfaces	P2S §3.2
$F(d)$	Per-channel cost function ($= d$, unique)	§5; P2S §1
γ_2/γ_1	Trace ratio ($= 17/4$, derived in Paper 4A)	§10
C_{total}	Total capacity ($= 61$)	§7; P2S §III.1
C_{vac}	Vacuum capacity ($= 42$)	§9; P2S §III.2
C_{mat}	Matter capacity ($= 19$)	§9
d_{eff}	Effective states per channel ($= (C_{\mathrm{total}} - 1) + C_{\mathrm{vac}} = 60 + 42 = 102$)	§9.4
s_{crit}	Critical saturation ($= 1/d_{\mathrm{eff}} = 1/102$)	§9
Ω_Λ	Vacuum density fraction ($= 42/61$)	§9
Ω_m	Matter density fraction ($= 19/61$)	§9
Q1, Q2	MECE partition predicates	§9
$E(g)$	Generation cost ladder ($g(g+1)/2$ in units of ε^*)	§11
N_{gen}	Number of generations ($= 3$, largest g with $E(g) \le C_{\mathrm{EW}}$)	§11
C_{EW}	Electroweak capacity budget for generations ($= \kappa \times \dim_{\mathrm{EW}} = 8$)	§11
L_{equip}	Horizon equipartition ($\Omega_{\mathrm{sector}} = \lvert\mathrm{sector}\rvert / C_{\mathrm{total}}$)	§9
$L_{\mathrm{hypercharge}}$	Unique hypercharge assignment ($z^2 - 2z - 8 = 0$)	§6
θ_{QCD}	QCD vacuum angle ($= 0$, derived from minimality)	§13.1
T_{proton}	Proton absolute stability (exact B conservation at saturation)	§13.3
$T_{\mathrm{vac,stab}}$	Vacuum absolute stability (unique admissibility well)	§13.2

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